NON-CONVENTIONAL ENERGY SOURCES

Subject Code: 23A03703a | UNIT – I : PRINCIPLES OF SOLAR RADIATION

I. Role & Potential of New and Renewable Sources

Energy sources are of two types: **Conventional** (coal, oil, gas, large hydro, nuclear) – limited stock, polluting; and **Non-conventional / renewable** (solar, wind, biomass, geothermal, ocean, small hydro) – replenished naturally, almost no pollution.

❖Why do we need renewable energy?

- Fossil fuel reserves are **finite** – coal ~100-150 yrs, oil/gas ~40-60 yrs at present rate.
- Burning fossil fuels causes CO_2, SO_2, NO_x emission → global warming, acid rain, smog.
- Energy demand rising fast with population + industrialisation. India imports ~85% of its crude oil.
- Renewables are **decentralised** – good for remote villages where the grid cannot reach.
- Energy security, rural employment, low running cost.

❖Potential in India

- Solar: ~5,000 trillion kWh/year incident on land; ~**300 clear sunny days** in most parts; 4–7 kWh/m^2/day.
- Wind: ~ 300 GW potential at 100 m hub height (Tamil Nadu, Gujarat, Karnataka, A.P., Maharashtra, Rajasthan).
- Biomass: ~ 25 GW (agri-residue, bagasse cogeneration), plus biogas from cattle dung.
- Small hydro: ~ 20 GW (< 25 MW stations).
- Geothermal: ~10,000 MW (Puga Valley – Ladakh, Manikaran – H.P., Tattapani – Chhattisgarh).
- National target: **500 GW non-fossil capacity by 2030**; Ministry: **MNRE**.

Renewable = inexhaustible + clean + local | Limitation = low energy density, intermittent, high initial cost

❖Comparison table

Feature	Conventional	Non-conventional
Availability	Limited	Unlimited
Pollution	High	Very low / nil
Initial cost	Moderate	High
Running cost	High (fuel)	Very low
Nature	Centralised	Mostly decentralised

2. The Solar Energy Option

- Sun = a giant fusion reactor, delivering ~ 1.8×10^{14} kW to earth; this is ~10,000 times mankind's present need.
- Free, widely distributed, clean, no moving fuel, long life of devices.
- **Forms of use:** (i) Thermal – heating, drying, cooking, power (ii) Photovoltaic – direct electricity (iii) Chemical – photosynthesis, solar hydrogen.
- **Disadvantages:** dilute (~1 kW/m^2 max), intermittent (night, clouds, season), needs storage/back-up, large land area, high capital cost.
- Solar radiation arrives as **short-wave** (0.3–3 µm) – ~7% UV, 47% visible, 46% IR.

3. Environmental Impact of Solar Power

❖Positive

- Zero emission during operation – no CO_2, SO_2, NO_x, no noise, no fuel transport.
- 1 kW rooftop PV saves ~ 1.2–1.5 tonnes CO_2 per year.
- Reduces dependence on fossil fuel; supports rural electrification.

❖Negative

- **Land use:** utility plants need ~ 2–3 hectares per MW; may displace habitat/agriculture.
- **Manufacturing:** PV cells use toxic chemicals (silicon tetrachloride, cadmium, arsenic, lead in solder); energy-intensive silicon purification.
- **Water use:** concentrating thermal plants need cooling water in arid zones.
- **Disposal:** end-of-life panels (25–30 yrs) → e-waste, recycling needed.
- **Heat island/ glare:** reflectors can harm birds (burn "streamers") and cause glare.

- Working fluids/ salts in solar ponds and storage may leak.

> Net environmental impact of solar is far lower than coal; energy pay-back time of a PV panel = 1–3 years

4. Physics of the Sun

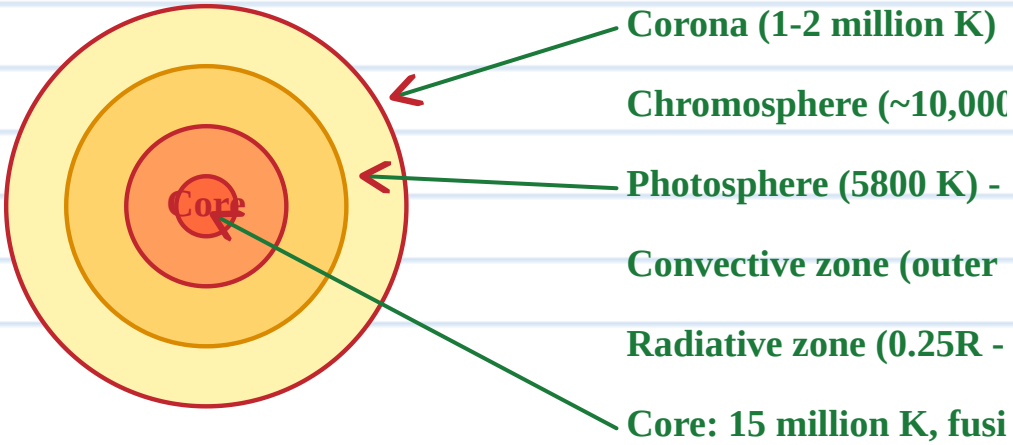


Fig 1.1 Structure of the Sun (not to scale)

- Diameter of sun = 1.39×10^9 m (about 109 × earth). Mass = 1.99×10^{30} kg. Mean distance from earth = 1.496×10^{11} m (1 AU).
- Effective black-body temperature = 5777 K (≈5800 K); core ≈ 15×10^6 K, density ≈ 150 g/cc.
- Made of ~ 73% H_2, 25% He, 2% heavier elements.

❖Energy source – nuclear fusion (proton-proton chain)

$$4\,{}^1H \rightarrow {}^4He + 2\,e^+ + 2\,\nu + 26.7\ MeV$$

- Mass lost ≈ 4.3×10^9 kg every second ($\Delta m \rightarrow E = mc^2$); total power output ≈ 3.8×10^{26} W.
- The energy diffuses outwards through the radiative and convective zones (takes ~ 10^5 yrs) and is emitted from the photosphere.

❖Spectral distribution

- Sun behaves like a black body at 5777 K – Wien's law: $\lambda_{max}\, T = 2898\ \mu m \cdot K$ → $\lambda_{max} \approx 0.50\ \mu m$ (green-yellow).
- Stefan–Boltzmann: $E = \sigma T^4$, $\sigma = 5.67 \times 10^{-8}\ W/m^2K^4$.

5. The Solar Constant (I_{sc})

> **Solar constant** = the energy from the sun, per unit time, received on a unit area of surface **perpendicular** to the rays, at the **mean earth-sun distance**, **outside the atmosphere.**

$$I_{sc} = 1367\ W/m^2\ (WMO\ value) = 1.94\ cal/cm^2/min = 4.92\ MJ/m^2h = 1.353\ kW/m^2\ (NASA)$$

❖Derivation (using Stefan-Boltzmann)

- Sun's power output $Q = 4\pi R_s^2 \sigma T_s^4$
- This is spread over a sphere of radius r (earth-sun distance), so
- $I_{sc} = (R_s / r)^2 \sigma T_s^4$
- With $R_s = 6.96\times10^8$ m, $r = 1.496\times10^{11}$ m, $T = 5777$ K → $I_{sc} \approx 1367\ W/m^2$.

❖Variation through the year

- Earth's orbit is elliptical → distance varies ±1.7%.
- Extra-terrestrial normal radiation on day n:

$$I_{on} = I_{sc}\,[\,1 + 0.033\cos(\,360\,n / 365\,)\,]$$

- Max ≈ 1412 W/m^2 (3 Jan, perihelion); Min ≈ 1322 W/m^2 (4 July, aphelion).

6. Extra-Terrestrial & Terrestrial Radiation

❖Extra-terrestrial radiation (ETR)

- Radiation falling on top of the atmosphere – undepleted; depends only on **latitude, day of the year, time** (since no atmosphere).
- Spectrum ≈ black-body 5777 K : UV 7%, visible 47%, IR 46%.
- On a **horizontal** surface at the top of atmosphere:

$$I_o = I_{on}\cos\theta_z = I_{on}\,(\sin\varphi\sin\delta + \cos\varphi\cos\delta\cos\omega)$$

- Daily total $H_o = (24/\pi)\,I_{sc}\,[1+0.033\cos(360n/365)]\,(\cos\varphi\cos\delta\sin\omega_s + (\pi\,\omega_s/180)\sin\varphi\sin\delta) \rightarrow J/m^2day$

❖Terrestrial radiation

- Radiation reaching earth's surface after passing the atmosphere; it is reduced and altered:
- Absorption by O_3 (UV), H_2O vapour & CO_2 (IR), dust.
- Scattering – by air molecules (Rayleigh, $\propto 1/\lambda^4$ → blue sky) and dust (Mie scattering).
- Reflection by clouds (albedo ≈ 30% on average for earth).
- Result: only ~ 70–75% reaches ground; peak ≈ 1000 W/m^2 at sea level on a clear day.

❖Components

- Beam (direct) radiation, I_b: received directly without scattering; casts shadow.
- Diffuse (sky) radiation, I_d: received after scattering by atmosphere; from all directions, no shadow.
- Global (total) radiation: $I = I_b \cos\theta + I_d$ (on a horizontal surface, $I = I_b \cos\theta_z + I_d$).
- Albedo/ reflected radiation from ground adds on tilted surfaces.

7. Sun–Earth Geometry & Angles

> Declination (δ): angle between sun's rays and the equatorial plane. Varies ±23.45° through the year.

$$\delta = 23.45 \sin[360\,(284 + n)/365]$$ (n = day number, 1 Jan = 1)

- δ = +23.45° on 21 June (summer solstice); 0° on 21 Mar & 23 Sep (equinox); −23.45° on 22 Dec.
- Latitude (φ): angular position north/south of equator (+ve north).
- Hour angle (ω): angular displacement of sun east/west of local meridian; 15° per hour; ω = 0 at solar noon, +ve afternoon, −ve morning.

$$\omega = 15\,(\text{Solar time} - 12)\ \text{degrees}$$

- Solar time = standard time ± 4 ($L_{st} - L_{loc}$) + E, where E = equation of time (min). 4 min per degree of longitude. For India, L_{st} = 82.5° E.
- Zenith angle θ_z, Altitude $\alpha = 90 - \theta_z$, Azimuth γ_s.

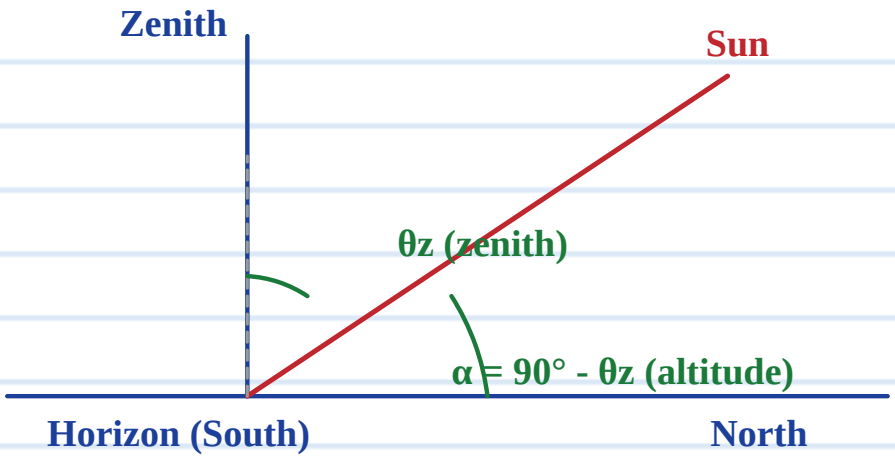


Fig 1.2 Zenith angle and solar altitude angle

$$\cos\theta_z = \sin\alpha = \sin\varphi \sin\delta + \cos\varphi \cos\delta \cos\omega$$

- Sunset hour angle: at sunset $\theta_z = 90° \rightarrow \cos\omega_s = -\tan\varphi \tan\delta$.
- Day length: $N = (2/15)\cos^{-1}(-\tan\varphi \tan\delta)$ hours.
- Surface azimuth (γ), slope (β), incidence angle (θ), and ground reflectance (ρ) are other terms.

❖Example

> Find the zenith angle at 10:00 solar time on 21 March at Hyderabad (φ = 17.4° N). $\delta = 0$, $\omega = 15(10-12) = -30°$. $\cos\theta_z = \cos17.4 \times \cos0 \times \cos(-30) = 0.954 \times 0.866 = 0.826 \rightarrow \theta_z = 34.3°$, altitude 55.7°.

8. Solar Radiation on a Tilted Surface

- Collectors are tilted (slope β) and face south (northern hemisphere), so we need radiation on the tilted plane from horizontal-plane data (I, I_d).

> Incidence angle θ = angle between the beam and the normal to the surface.

$$\cos\theta = \sin\delta \sin\varphi \cos\beta - \sin\delta \cos\varphi \sin\beta \cos\gamma + \cos\delta \cos\varphi \cos\beta \cos\omega + \cos\delta \sin\varphi \sin\beta \cos\gamma \cos\omega + \cos\delta \sin\beta \sin\gamma \sin\omega$$

- For a south-facing surface ($\gamma = 0$) this reduces to:

$$\cos\theta = \sin\delta \sin(\varphi - \beta) + \cos\delta \cos\omega \cos(\varphi - \beta)$$

- For a horizontal surface $\beta = 0 \rightarrow \cos\theta = \cos\theta_z$.
- For a surface that always faces the sun (tracking) $\rightarrow \theta = 0$.

❖Tilt factor (R_b)

$$R_b = I_{bt} / I_{bh} = \cos\theta / \cos\theta_z$$

❖Liu & Jordan isotropic-sky model

$$I_T = I_b R_b + I_d (1 + \cos\beta)/2 + \rho (I_b + I_d)(1 - \cos\beta)/2$$

- Term 1 = beam on tilted plane; Term 2 = diffuse from sky (sky view factor $(1+\cos\beta)/2$); Term 3 = ground-reflected ($\rho \approx 0.2$ for ground, 0.7 for snow).
- Best fixed tilt: $\beta \approx$ latitude (φ) for annual use; $\varphi + 15°$ for winter; $\varphi - 15°$ for summer.
- Monthly average: $\bar{R} = (1 - \bar{H_d}/\bar{H})\,\bar{R_b} + (\bar{H_d}/\bar{H})(1+\cos\beta)/2 + \rho(1-\cos\beta)/2$.

❖Example

Hyderabad ($\varphi = 17.4°$), noon ($\omega = 0$) on 21 June ($\delta = 23.45°$), south-facing surface tilted $\beta = 20°$. $\cos\theta = \sin 23.45 \sin(17.4-20) + \cos 23.45 \cos(-2.6) = 0.398\times(-0.0454) + 0.917\times0.999 = 0.898 \rightarrow \theta \approx 26.1°$.

9. Instruments for Measuring Solar Radiation

Radiometer → measures radiation (W/m^2). **Pyranometer** → global (beam+diffuse) on a plane. **Pyrheliometer** → beam (direct normal) only.

❖A. Pyranometer

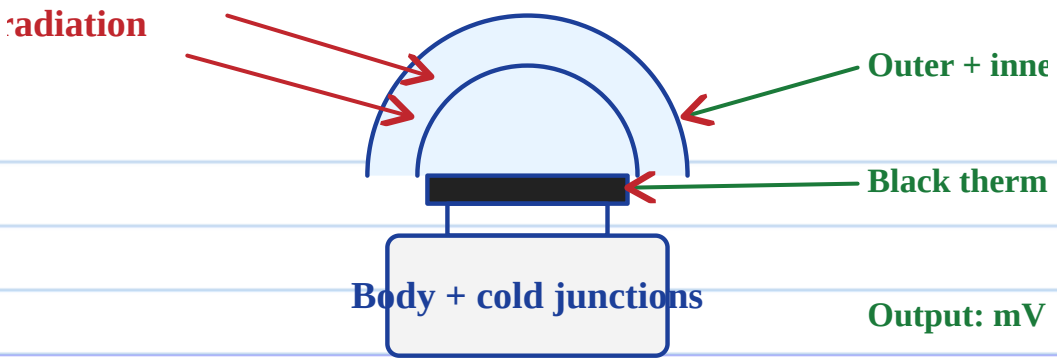


Fig 1.3 Pyranometer (thermopile type) – measures global (beam + diffuse) radiation

- **Thermopile type (Eppley / Kipp & Zonen):** a black, blackened receiver heated by radiation; temperature difference between hot (black) and cold (white) junctions → emf (mV) ∝ radiation. Covered by two hemispherical glass domes (0.3–3 μm transmission) to cut convective loss. Silica gel dryer inside.

- **Photovoltaic type:** silicon cell short-circuit current ∝ radiation – cheap but not spectrally flat.
- **Shadow-ring (shading band) method:** a band shades the pyranometer from direct sun → measures diffuse only. Correction factor applied.
- **Accuracy:** ± 1–3%; calibrated against a reference pyrheliometer.

❖B. Pyrheliometer (beam radiation)

- A long collimating tube with small field of view (~5.7°) → only direct sun disc + circumsolar region enters.
- Must be kept **pointed at the sun** with a **tracker** (equatorial mount).
- Types: **Angstrom compensation pyrheliometer** (two strips; one shaded & electrically heated until temperatures equal – electrical power = radiation), **Abbot silver-disc, Eppley normal incidence pyrheliometer (NIP).**
- Thermopile output → recorder.

❖C. Pyrgeometer / Net pyrradiometer

- Pyrgeometer measures long-wave (infrared) sky/earth radiation (4–100 μm).
- **Net pyrradiometer** – two blackened receivers facing up and down → net all-wave radiation.

10. Instruments for Sunshine Duration

> **Sunshine hours** = number of hours during which direct sunlight is strong enough (> 120 W/m^2 – WMO) to burn the record. **Bright sunshine** ≈ 0.2 kW/m^2 threshold.

❖Campbell–Stokes sunshine recorder

- A solid **glass sphere (10 cm)** acts as a lens, focusing sun rays on a **card strip** held in a spherical bowl behind it.
- The image burns a trace on the card as sun moves; **length of burn = sunshine duration.**
- Three different cards used: long curved (summer), short curved (equinox), straight (winter).
- Simple, cheap, no power. Limitation: thin cloud/ low sun is hard to record; observer needs to read the card daily.

❖ **Jordan sunshine recorder**

- Uses a semi-cylindrical box with two slits; light falls on **photosensitive paper** (blue-print) inside – the exposed line shows duration.

❖ **Modern: Pyrheliometric / photoelectric**

- Compares the beam signal with 120 W/m^2 threshold electronically → logs sunshine hours.

❖ **Uses of sunshine data**

- Gives **ratio n/N** (n = actual bright sun hours, N = max possible day length).
- Used in **Angstrom-Prescott equation** to estimate global radiation where no pyranometer is available:

$$\bar{H} / \bar{H_o} = a + b\,(\bar{n} / \bar{N})$$

- a, b = regression constants (typical $a \approx 0.25$, $b \approx 0.50$; for India often $a=0.27$, $b=0.47$).
- $\bar{H}$ = monthly mean daily global radiation on horizontal surface; $\bar{H_o}$ = extra-terrestrial.
- **Clearness index** $\bar{K_T} = \bar{H} / \bar{H_o}$.

II. Solar Radiation Data

- Needed for **designing and sizing** solar systems (collector area, PV array size, storage).
- **Sources:** (i) India Meteorological Department (IMD) network – ~45 stations measuring global + diffuse; (ii) **NREL / MNRE – Solar Radiation Handbook & Solar Resource Assessment (SRRA) stations;** (iii) NASA, PVGIS, Meteonorm (satellite derived).
- **Typical data given:** monthly average daily global radiation $\bar{H}$ (MJ)/m^2 day), diffuse $\bar{H_d}$, sunshine hours, ambient temperature.
- **Data types:** hourly, daily, monthly mean, **Typical Meteorological Year (TMY)** files.
- **Satellite methods** (INSAT, Kalpana) estimate radiation from cloud cover.

❖Average daily global radiation in India

Region	kWh/m²/day
Rajasthan, Gujarat (W. India)	5.5 – 6.5
Deccan plateau (A.P., Karnataka)	5.0 – 5.8
Eastern coast	4.5 – 5.5
North-East, Kerala (monsoon)	4.0 – 4.7

- India receives ~ 4–7 kWh/m²/day and 2300–3200 sunshine hours/yr.
- Units: 1 kWh/m² = 3.6 MJ/m²; 1 Langley = 1 cal/cm² = 41.87 kJ/m²; 1 W/m² = 0.86 kcal/m²h.

12. Solved Problems

❖Problem 1 – Day length

> Find the day length at $\varphi = 28^\circ$ N on 21 June. ($\delta = 23.45^\circ$)

$\cos \omega_s = -\tan 28^\circ \times \tan 23.45^\circ = -0.5317 \times 0.4336 = -0.2305 \rightarrow \omega_s = 103.3^\circ \rightarrow N = 2\times103.3/15 = 13.77$ h.

❖Problem 2 – Declination

> Declination on 15 April (n = 105): $\delta = 23.45 \sin[360(284+105)/365] = 23.45 \sin(383.7^\circ) = 23.45 \sin(23.7^\circ) = 9.4^\circ$.

❖Problem 3 – Extra-terrestrial normal radiation on 1 Feb (n = 32)

> $I_{on} = 1367[1 + 0.033 \cos(360\times32/365)] = 1367[1 + 0.033\times\cos 31.56^\circ] = 1367\times1.0281 = 1405$ W/m².

❖Problem 4 – Solar time

> Standard time 10:30 IST at Hyderabad (longitude 78.5° E) on a day when E = −10 min. Solar time = 10:30 + 4(78.5 − 82.5) min + E = 10:30 − 16 min − 10 min = 10:04.

❖Problem 5 – Angstrom

> $\bar{n}/N = 0.7$, $H_o = 36$ MJ/m²d, a = 0.25, b = 0.50 → $H = 36\,(0.25 + 0.5\times0.7) = 36 \times 0.60 = 21.6$ MJ/m²d.

13. Important Questions (Exam Oriented)

- Explain the role and potential of new & renewable energy sources.

- What are the environmental impacts of solar power?
- Define the solar constant; derive it, and explain the variation of ETR through the year.
- Define declination, hour angle, zenith angle, incidence angle. Derive $\cos\theta_z$.
- Explain terrestrial solar radiation; beam, diffuse and global components.
- Derive the expression for radiation on a tilted surface (Liu & Jordan).
- With a neat sketch, explain the pyranometer and pyrheliometer.
- Explain the Campbell–Stokes sunshine recorder.
- Write notes on solar radiation data & Angstrom correlation.

Tip: draw neat labelled diagrams, write equations clearly with units – full marks!

14. Two More Problems

❖Problem 6 – Sunset hour angle & sunrise time

At $\varphi = 17.4°$ N on 22 Dec ($\delta = -23.45°$): $\cos\omega_s = -\tan 17.4 \times \tan(-23.45) = +0.3134 \times 0.4336 = 0.1359 \rightarrow \omega_s = 82.2° \rightarrow$ day length $= 2 \times 82.2/15 = 10.96$ h; sunrise ≈ 06:31 solar time, sunset ≈ 17:29 solar time.

❖Problem 7 – Radiation on tilted plane

$I_b = 600$, $I_d = 120$ W/m² (horizontal basis, $I_{bh} = 600$), $R_b = 1.2$, $\beta = 30°$, $\rho = 0.2$.

$I_T = 600(1.2) + 120(1+0.866)/2 + 0.2(720)(1-0.866)/2 = 720 + 112 + 9.6 = 841.6$ W/m².

15. Quick Revision – Formula Sheet

Quantity	Formula
Solar constant	$I_{sc} = 1367\ W/m^2$
ETR on day n	$I_{on} = I_{sc}[1+0.033\cos(360n/365)]$
Declination	$\delta = 23.45 \sin[360(284+n)/365]$
Hour angle	$\omega = 15(t_{solar} - 12)$
Zenith	$\cos\theta_z = \sin\varphi \sin\delta + \cos\varphi \cos\delta \cos\omega$
Sunset angle	$\cos\omega_s = -\tan\varphi \tan\delta$
Day length	$N = (2/15)\,\omega_s$ hours
Tilted incidence ($\gamma=0$)	$\cos\theta = \sin\delta \sin(\varphi-\beta) + \cos\delta \cos\omega \cos(\varphi-\beta)$
Tilt factor	$R_b = \cos\theta / \cos\theta_z$
Angstrom	$H/H_o = a + b(n/N)$

❖Key terms in one line each

- **Air mass (m):** ratio of path length of beam through atmosphere to the path when sun is at zenith; $m = 1/\cos\theta_z = 1$ at zenith, AM 1.5 standard for PV testing.
- **Sky clearness / Turbidity:** measure of dust and water vapour in atmosphere.
- **Insolation:** incident solar radiation energy per unit area (Wh/m^2).
  Irradiance: power per unit area (W/m^2).

NON-CONVENTIONAL ENERGY SOURCES

Subject Code: 23A03703a | UNIT – II : SOLAR ENERGY COLLECTION

1. Introduction – Solar Collectors

A solar collector is a special heat exchanger that absorbs solar radiation, converts it to heat and transfers the heat to a working fluid (water, air, oil).

❖Classification

- Non-concentrating (stationary): aperture area = absorber area → flat plate collector (FPC), evacuated tube collector (ETC). Gives low-medium temperature (40–100 °C).
- Concentrating (focusing): aperture area > absorber area → reflectors/ lenses focus rays. Gives 100–2000 °C.
- By fluid: liquid collectors (water, oil), air collectors.
- By tracking: non-tracking, single-axis, two-axis.
- Concentration ratio $C = A_a / A_r$ (aperture area / receiver area).

2. Flat Plate Collector (FPC)

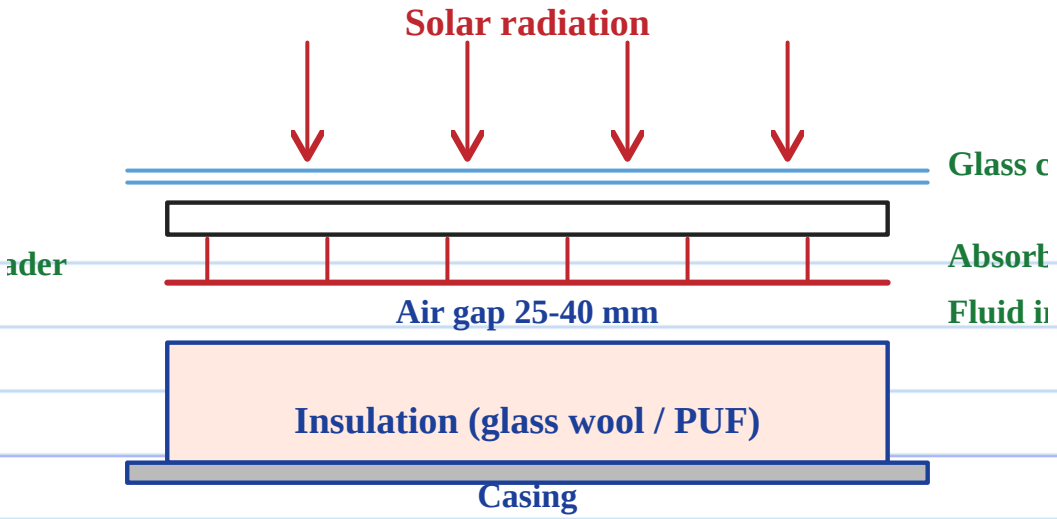


Fig 2.1 Cross-section of a liquid flat plate collector

❖Components

- Absorber plate: copper / aluminium / steel sheet, blackened with selective coating (e.g., black chrome, black nickel) – high absorptance α and low emittance ε.

- **Tubes (risers) and headers:** tubes welded/brazed to the plate carry the fluid; pitch 10–15 cm, tube dia 1–1.5 cm.
- **Transparent cover:** one or two glass sheets (3–5 mm) – transmits short wave radiation, **traps long-wave re-radiation** (greenhouse effect); also cuts convection loss.
- **Insulation:** glass wool / mineral wool / PUF, 5–10 cm at back and sides.
- **Casing (box):** aluminium/GI/FRP; weather-proof.

❖Working

- Solar radiation passes through glass, strikes the black absorber plate and is converted to heat; heat flows to the tubes and into the fluid by conduction and convection.
- Thermo-siphon (natural circulation) or pump-driven circulation to a storage tank.
- Tilt = latitude; faces south; efficiency 40–60%; fluid temp up to ~ 80–90 °C; stagnation ~ 150 °C.

❖Advantages & limitations

- Absorbs **both beam & diffuse** radiation; no tracking; simple, cheap, low maintenance.
- Limitations: low temperature, large heat loss at high ΔT, efficiency drops on cloudy days.

❖Selective surface

> **Selective surface:** high α for short-wave (0.3–3 μm), low ε for long-wave (> 3 μm) so it absorbs well yet radiates poorly. Ratio $\alpha/\varepsilon > 10$ (e.g. black chrome $\alpha = 0.95$, $\varepsilon = 0.1$).

3. Thermal Analysis of Flat Plate Collector

❖Energy balance

- Radiation absorbed (per unit area) $S = I_T (\tau\alpha)$ where τ = transmittance of cover, α = absorptance of plate, I_T = radiation on tilted surface.
- Heat loss = $U_L (T_p - T_a)$

$$\text{Useful heat gain } Q_u = A_c [S - U_L (T_p - T_a)]$$

- Also $Q_u = \dot{m}\, C_p (T_{fo} - T_{fi})$.

❖Overall loss coefficient $U_L = U_t + U_b + U_e$

- U_t top loss (convection + radiation through the glass) ≈ 3–8 W/m^2K
- U_b bottom loss = k/L of insulation ≈ 0.5 W/m^2K
- U_e edge loss – small.

❖Collector efficiency factor F' and heat removal factor F_R

- Since plate temperature varies along the flow, we relate gain to **inlet fluid temp** instead:

$$Q_u = A_c F_R [S - U_L (T_{fi} - T_a)]$$

- $F_R = \dot{m} C_p / (A_c U_L) \times [1 - \exp(-A_c U_L F' / \dot{m} C_p)]$
- F_R = (actual useful gain) / (gain if whole plate were at inlet fluid temp). $0 < F_R < 1$.

❖Instantaneous efficiency (Hottel–Whillier–Bliss equation)

$$\eta = Q_u / (A_c I_T) = F_R(\tau\alpha) - F_R U_L (T_{fi} - T_a) / I_T$$

- Plot η vs $(T_{fi} - T_a)/I_T$ → **straight line**: intercept = $F_R(\tau\alpha)$ (optical efficiency), slope = $-F_R U_L$ (loss).
- Higher intercept & lower slope → better collector. Used in **testing** (ASHRAE 93 / BIS).

❖Fin efficiency (standard fin of the tube-plate)

- Plate between tubes acts as a fin: $F = \tanh[m(W-D)/2] / [m(W-D)/2]$, $m = \sqrt{(U_L/k\delta)}$. W = tube spacing, D = tube dia, δ = plate thickness.

❖Example

> FPC area 2 m^2, I_T = 800 W/m^2, $(\tau\alpha)$ = 0.8, U_L = 6 W/m^2K, F_R = 0.85, T_{fi} = 40 °C, T_a = 25 °C.

$Q_u = 2 \times 0.85[800 \times 0.8 - 6 \times 15] = 1.7 \times 550 = 935$ W, $\eta = 935/(2 \times 800) = 58.4$ %.

4. Evacuated Tube Collector (ETC)

- Absorber tube inside a **vacuum-sealed glass tube** → **no convective loss**; uses heat-pipe or U-tube; efficient in cold, cloudy climates; up to 120–150 °C.
- Glass-glass (Dewar type): two concentric tubes, vacuum in between, selective coating on the inner tube.

- Costlier than FPC; fragile; but low $U_L \approx 1\text{–}2\ W/m^2K$.

5. Solar Air Heater

- Same as FPC but fluid is **air** (low C_p, low h → needs large area / fins / roughness / corrugated surface).
- Types: bare plate, **finned**, **corrugated**, **V-corrugated**, matrix-packed, **with artificial roughness**, double-pass.
- Uses: crop drying, space heating, seasoning of timber. No freezing / leakage / corrosion issues.

6. Concentrating Collectors

> A concentrating collector uses **reflectors or lenses** to focus beam radiation on a small **receiver**, hence high temperature and small heat loss area.

Concentration ratio $C = A_a / A_r$ (aperture area / absorber area)

- **Maximum possible C (ideal, 2-D):** $C_{max} = 1 / \sin \theta_a$ (θ_a = acceptance half-angle). For 3-D (point focus): $C_{max} = 1 / \sin^2 \theta_a$. The sun's half-angle = $0.267°$ → $C_{max} \approx 215$ (2-D) & 46,000 (3-D).
- **Acceptance angle ($2\theta_a$):** range of incidence angles within which all rays reach the receiver without tracking. Larger C ⇒ smaller acceptance angle ⇒ accurate tracking needed.
- They use **only beam radiation** (diffuse is lost) → need clear sky.

❖Advantages vs flat plate

- Higher temperature, higher efficiency at high temperature, less heat-loss area, less costly absorber material, can use thermodynamic cycles (power generation).

❖Disadvantages

- Need sun tracking, lose diffuse radiation, reflector surfaces get dirty/ need cleaning, high cost, complex.

7. Classification of Concentrating Collectors

Basis	Types
Focus	Line focus (trough) / Point focus (dish, tower)
Reflector	Reflecting (mirror) / Refracting (Fresnel lens)
Receiver	Convex, flat, cylindrical, cavity
Tracking	Single-axis / Two-axis / Non-tracking (CPC)
Contact	Imaging / Non-imaging

❖(a) Parabolic trough collector (PTC) – line focus

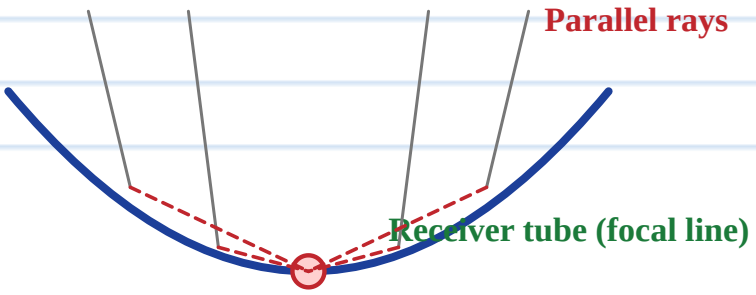


Fig 2.2 Parabolic trough concentrator (line focus)

- Parabolic cylinder mirror (aluminium/ silvered glass) with a **receiver tube** (stainless steel with selective coating inside an evacuated glass envelope) at the focal line.
- Heat transfer fluid: synthetic oil (Therminol) → 390 °C; steam generation (DSG).
- **One-axis tracking** (E–W or N–S axis). C = 15–80. Used in SEGS plants in California (354 MW) and Indian plants at Gurgaon/Rajasthan.
- **Parabola equation:** $y^2 = 4 f x$; focal length f. Rim angle ψ_r: $f / a = (1 + \cos \psi_r) / (4 \sin \psi_r)$ (a = half aperture).

❖(b) Parabolic dish (point focus)

- Paraboloid dish (diameter 3–25 m) with receiver at the focal point; two-axis tracking; C = 500–3000; T up to 1500 °C.
- Usually coupled with a **Stirling engine** – Dish-Stirling (25 kW), efficiency ~ 30% (highest of all solar-to-electric). Also used for solar cookers (Scheffler dish, Parabolic community cooker).

❖(c) Central receiver / Power tower (heliostat field)

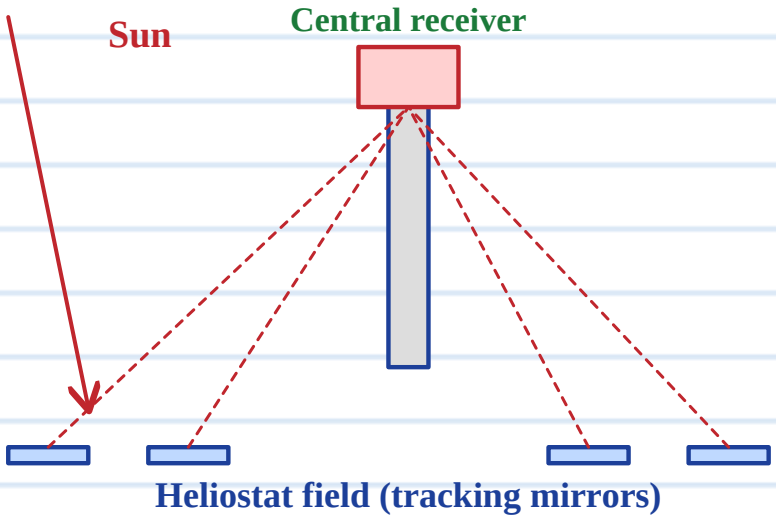


Fig 2.3 Central receiver (power tower) system

- **Heliostats** – large flat/slightly curved mirrors with two-axis tracking that all reflect sun on one **central receiver** on a tower. C = 300–1500, T = 500–1000 °C.
- Fluid: molten salt (60% $NaNO_3$ + 40% KNO_3), steam, air → storage → turbine. e.g. Solar Two, Gemasolar (Spain), Ivanpah (USA).

❖(d) Fresnel lens collector

- A thin flat plastic **Fresnel lens** (concentric grooved rings) refracts the rays to a focal point/line; low weight, low cost. Linear Fresnel reflector (LFR): long flat mirror strips on a single axis with a fixed elevated receiver.

❖(e) Compound parabolic concentrator (CPC) – Winston collector

- **Non-imaging**; two parabolic segments with different focal axes; accepts radiation over a wide acceptance angle (± 2–6°); needs **only seasonal tilt adjustment**, no daily tracking; C = 3–10; T up to 200 °C. Collects some diffuse radiation too.

8. Orientation & Tracking

- **Fixed FPC**: face south (N-hemisphere), tilt ≈ latitude (+ 10–15° for winter use). Seasonal manual adjustment gives ~ 5–10% gain.
- **Tracking modes**
  - **Polar (equatorial) mount**: one axis parallel to earth's axis, rotates 15°/h; keeps θ small.
  - **N–S horizontal axis, E–W tracking**: good in summer, lower in winter.
  - **E–W horizontal axis, N–S tracking**.
  - **Two-axis (azimuth-altitude)**: θ ≈ 0 always; best but costly.

- Trackers: mechanical (clock drive), sensor/ photo-cell based, microprocessor/ astronomical program.
- Optical errors: surface slope error, tracking error, mirror reflectivity, receiver mis-alignment → intercept factor γ.

9. Thermal Analysis of Concentrating Collectors

❖Optical efficiency

$$\eta_o = \rho \cdot \tau \cdot \alpha \cdot \gamma \text{ (reflectance × transmittance × absorptance × intercept factor)}$$

- Typical η_o = 0.70 – 0.85 for a good parabolic trough.

❖Energy balance on the receiver

- Absorbed flux $S = I_b \eta_o$ (W per m^2 of aperture, with I_b = beam radiation on aperture, including cosine factor).
- Heat loss from receiver = $U_L A_r (T_r - T_a)$.

$$Q_u = A_a S - A_r U_L (T_r - T_a) = \dot{m} C_p (T_{fo} - T_{fi})$$

- Using F_R: $Q_u = F_R [A_a S - A_r U_L (T_{fi} - T_a)]$
- Thermal efficiency:

$$\eta = Q_u / (A_a I_b) = F_R [\eta_o - U_L (T_{fi} - T_a) / (C \cdot I_b)]$$

- Key insight: the loss term is divided by the concentration ratio C → at high C, losses are small, so efficiency stays high even at high temperatures.
- Compare FPC: $\eta = F_R[(\tau\alpha) - U_L(T_{fi}-T_a)/I_T]$ (C = 1).

❖Example

> PTC: aperture 2.5 m × 10 m (A_a = 25 m^2), receiver dia 0.05 m (A_r = $\pi \times 0.05 \times 10$ = 1.57 m^2), I_b = 700, η_o = 0.75, U_L = 8, F_R = 0.9, T_{fi} = 150 °C, T_a = 30 °C.

C = 25/1.57 = 15.9. Q_u = 0.9[25×700×0.75 – 1.57×8×120] = 0.9[13125 – 1507] = **10,456 W**; η = 10456/17500 = **59.7 %**.

10. Advanced Collectors

❖(i) Evacuated tube collectors – heat-pipe type

- A sealed copper **heat pipe** with a volatile fluid (alcohol/water) inside each evacuated tube; evaporation at hot end, condensation at the manifold bulb → very fast heat transfer, can't freeze/leak. Used for hot water at 60–120 °C.

❖(ii) Compound parabolic concentrator (CPC) + evacuated tube

- Combines low heat loss with moderate concentration, no tracking.

❖(iii) Solar chimney / transpired collector

- Perforated dark metal sheet on a wall; air sucked through holes is pre-heated – efficiency 60–80% for ventilation air.

❖(iv) Photovoltaic–thermal (PV/T) hybrid collector

- A PV module with fluid tubes at the back: produces electricity **and** heat; cooling raises PV efficiency; overall efficiency ~ 60–70%.

❖(v) Integrated collector-storage system (ICS)

- The tank itself is the absorber ("batch heater") – cheap but heat loss at night.

❖(vi) Solar tracking mirror (Scheffler) / Fresnel dish

- Fixed focus, used in community kitchens (Shirdi, Mount Abu).

❖(vii) Cavity receivers and volumetric receivers

- Cavity (box) receivers trap radiation (blackbody effect) – very low re-radiation; ceramic **volumetric** receivers heat air up to 1000 °C.

❖(viii) Nanofluid / direct-absorption collectors

- Nano-particles (Al_2O_3, CuO, carbon) suspended in fluid absorb radiation directly → higher efficiency; research stage.

❖(ix) Solar tower + molten salt storage

- 24-hour dispatchable power: salts stored at 565 °C in hot tank, used on demand.

11. Comparison – FPC vs Concentrating Collector

Feature	Flat plate	Concentrating
Radiation used	Beam + diffuse	Beam only
Tracking	Not needed	Needed (except CPC)
Temperature	< 100 °C	100 – 2000 °C
Concentration ratio	1	> 1 (up to 46,000)
Cost	Low	High
Heat loss area	Large (= aperture)	Small
Maintenance	Simple	Complex (cleaning, tracking)
Uses	Hot water, heating	Power, process heat, furnaces

12. Collector Performance Terms

- **Collector efficiency** = useful energy / incident energy (instantaneous or daily).
- **Stagnation temperature:** equilibrium temperature when no fluid flows (loss = absorbed). $T_{stg} = T_a + S/U_L$.
- **Thermal capacity & response time:** time constant = $(mC)/(A\,U_L)$.
- **Incidence angle modifier** $K_{\tau\alpha} = (\tau\alpha)/(\tau\alpha)_n = 1 - b_o(1/\cos\theta - 1)$.
- **Effect of number of covers:** 1 cover – more transmittance but more loss; 2 covers used above 60 °C.
- **Effect of dust, wind & fluid flow rate** on performance.

13. Solved Problems

❖Problem 1 – Concentration ratio

A parabolic trough has aperture width 3 m and receiver diameter 6 cm. C (area basis) = 3 / (π×0.06) = 15.9.

❖Problem 2 – Maximum concentration

For a 2-D concentrator with $\theta_a = 0.267°$: $C_{max} = 1/\sin 0.267° = 1/0.00466 = 214.6$.

❖Problem 3 – Stagnation temperature

$S = 700\ W/m^2$, $U_L = 5$, $T_a = 30°C \rightarrow T_{stg} = 30 + 700/5 = 170\ °C$.

❖Problem 4 – Flow rate for given temperature rise

> Q_u = 1500 W, ΔT = 10 K, C_p = 4186 → $\dot{m}$ = 1500/(4186×10) = **0.0358 kg/s (129 kg/h).**

14. Important Questions

- Describe the construction and working of a liquid flat plate collector with neat sketch.
- Derive the Hottel-Whillier-Bliss equation; define F_R, F', U_L.
- How are concentrating collectors classified? Explain parabolic trough, dish & central receiver.
- Define concentration ratio & acceptance angle; show $C_{max} = 1/\sin\theta_a$.
- Explain CPC and its advantages. Write about thermal analysis of cylindrical parabolic collector.
- Write short notes on advanced collectors and orientation/ tracking.

> Remember: "Concentrating collectors ⇒ beam radiation only ⇒ tracking needed"

15. Solar Water Heating Systems (Application of FPC)

❖Natural circulation (thermosiphon) system

- Tank placed **above** the collector; hot water (lighter) rises into the tank, cold water descends to the collector bottom → no pump. Typical 100–200 LPD domestic system with 2 m^2 collector.
- Pressure head $\Delta p = (\rho_c - \rho_h)\, g\, H$ drives the flow.

❖Forced circulation system

- Pump + differential temperature controller; used for large systems (hospitals, hotels); tank can be placed anywhere.
- **Direct** (water in the collector) vs **indirect** (anti-freeze fluid with heat exchanger – for cold regions).

❖Design hints

- 1 m^2 of FPC heats ~ 50–60 litres/day to 60 °C; collector tilt ≈ latitude; avoid shade; insulate pipes; fit an **electric back-up** heater.
- Hard water scaling → use soft water or indirect loop.

16. Solar Cooker (Concentrating application)

- **Box type:** insulated box, glass lid, reflector mirror; 100–140 °C; slow cooking of rice/dal in 2–3 h. Efficiency ~ 25–30%.
- **Parabolic dish cooker:** reaches 250–300 °C, fast cooking, needs tracking.
- **Community/ Scheffler cooker:** steam cooking for 1000s of meals.
- Figure of merit: $F_1 = (T_p - T_a)/H_I$ (BIS test) – should be ≥ 0.12 m^2K/W.

17. Testing of Collectors (short note)

- Steady-state test: hold T_{fi} constant, measure I_T, $\dot{m}$, T_{fo} for several T_{fi} values; clear sky, near-normal incidence, wind 2–4 m/s.
- Plot η vs $(T_{fi} - T_a)/I_T$ → intercept = $F_R(\tau\alpha)$, slope = $-F_RU_L$.
- Also: time-constant test, incident-angle-modifier test, stagnation test, pressure-drop test, thermal-shock test.

18. Quick Revision Table

Term	Remember
FPC efficiency	$\eta = F_R(\tau\alpha) - F_RU_L\Delta T/I_T$
Useful heat	$Q_u = A\,F_R[S - U_L(T_{fi}-T_a)]$
Concentration ratio	$C = A_a/A_r$
C_{max} (2D / 3D)	$1/\sin\theta_a$; $1/\sin^2\theta_a$
Optical efficiency	$\rho\ \tau\ \alpha\ \gamma$
Stagnation temp	$T_a + S/U_L$

19. Worked Example – Number of Collectors for a Hostel

Need 1000 litres of hot water per day, raised from 25 °C to 60 °C. Energy = 1000×4.186×35 = **146.5 MJ/day**. Daily radiation on tilted plane = 18 MJ/m^2, average daily efficiency = 45 % → useful gain = 8.1 MJ/m^2day. Area = 146.5/8.1 = **18.1 m^2** ≈ 9 collectors of 2 m^2.

20. Sketch Tips for the Exam

- Always show the **sun rays**, label every layer (cover, air gap, absorber, insulation), mark flow direction **in / out**.

- For concentrators draw the **focal point**, aperture W, receiver and rim angle ψ_r.
- Write the governing equation beneath the diagram.

NON-CONVENTIONAL ENERGY SOURCES

Subject Code: 23A03703a | UNIT – III : SOLAR STORAGE & APPLICATIONS, WIND ENERGY

1. Need for Energy Storage

- Solar energy is **intermittent** (night, clouds, seasons) while demand continues → storage matches supply and demand.
- Storage types: **thermal, electrical (battery), chemical (hydrogen), mechanical (pumped hydro, flywheel, compressed air).**
- **Desirable features:** high energy density, low cost, long life, low loss, easy charge/ discharge, safe.

2. Sensible Heat Storage

> Heat stored by **raising the temperature** of a medium without change of phase.

$$Q = m\,C_p\,(T_f - T_i) = \rho\,V\,C_p\,\Delta T$$

- **Liquid media:** water (C_p = 4.18 kJ/kgK – cheap, best up to 100 °C), oil (to 300 °C), molten salts (to 565 °C), liquid sodium.
- **Solid media:** rock bed/ pebble bed (air collectors, $C_p \approx 0.85$ kJ/kgK), concrete, bricks, cast iron.
- Storage capacity depends on **mass, specific heat and ΔT**; water tank is the most common – **insulate** to cut loss.
- **Advantages:** simple, cheap, no phase issues. **Disadvantages:** large volume, temperature falls as heat is withdrawn, heat losses.

❖Rock-bed storage

- Air flows through packed bed of 2–5 cm pebbles; **charge** hot air top → bottom, **discharge** reverse (to keep stratification); used with solar air heaters for crop dryers and space heating.

3. Latent Heat Storage (PCM)

> Heat stored/ released during **phase change** (melting–solidification) at nearly **constant temperature**.

$$Q = m\,[\,C_{sp}(T_m - T_i) + a_m\, h_{sf} + C_{lp}(T_f - T_m)\,]$$

- (solid sensible + latent of fusion + liquid sensible); a_m = fraction melted, h_{sf} = latent heat.
- **Phase change materials (PCMs):**

PCM	Melting point °C	Latent heat kJ/kg
Paraffin wax	50 – 60	200 – 250
Glauber salt $Na_2SO_4{\cdot}10H_2O$	32	250
Calcium chloride hexahydrate	29	190
Fatty acids / stearic acid	55 – 70	180
$NaNO_3$ / KNO_3 salts	220 – 300	100 – 170

- **Advantages:** 5–14 times more energy per unit volume than sensible storage; nearly constant T delivery.
- **Problems:** low thermal conductivity (add fins/ graphite), **supercooling**, phase segregation/ incongruent melting (hydrated salts), corrosion, cost. Encapsulation in tubes/ spheres/ macro-capsules.
- **Uses:** solar water heaters, cooling load shifting, **building walls**, cooking at night.

4. Stratified Storage

- In a vertical **water tank**, hot water (low density) naturally stays at top and cold at bottom with a thin **thermocline** between them.
- Benefits: collector gets **cold water** at the bottom (higher efficiency); consumer gets **hot water** from the top; delivery temperature stays high.
- Keep inlet velocity low (diffusers, baffles) to avoid mixing; tall slim tank (H/D $\geq$ 2–3).
- **Fully mixed vs stratified:** stratified gives ~ 10–20% higher system performance.

5. Thermo-chemical & other storage (brief)

- **Chemical storage:** reversible reactions, e.g. $CaO + H_2O \rightleftharpoons Ca(OH)_2$ + heat; ammoniates; metal hydrides; **solar hydrogen** by electrolysis.
- **Electrical:** lead-acid, Li-ion, flow batteries for PV.
- **Mechanical:** pumped hydro, flywheel, compressed air.
- **Comparison:** Sensible (cheap, bulky) < Latent (compact, constant T) < Chemical (very compact, long-term, complex).

6. Solar Ponds

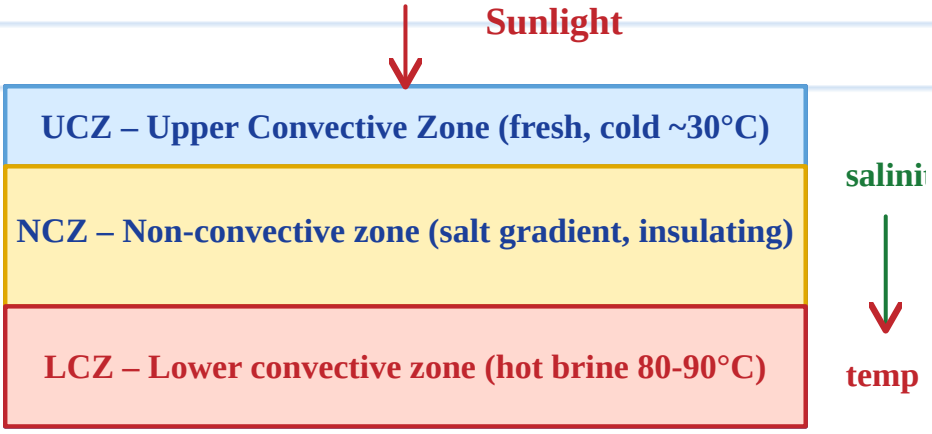


Fig 3.1 Salt-gradient solar pond

> A solar pond is a shallow body of salty water that **collects and stores solar energy** as heat at the bottom without any collector.

❖Three zones

- **UCZ (upper convective zone):** thin (0.1–0.5 m), fresh/ low salinity, near ambient temp, wind-mixed.
- **NCZ (non-convective / gradient zone):** 1–1.5 m, salinity increases with depth → **density gradient suppresses convection;** acts as a **transparent insulator.**
- **LCZ (lower convective zone / storage zone):** 1–2 m, **saturated brine,** hot (70–90 °C), well-mixed, black bottom lining (butyl/ HDPE).

❖Working principle

- **Normal water:** heated water rises → loses heat at top. In a solar pond heated bottom brine is **still denser** than the upper layer → stays at bottom; heat can't escape by convection; only conducts (water poor conductor).

- Heat extracted by pumping hot brine through an external heat exchanger (or an organic Rankine engine).

❖Stability criterion

> $\partial\rho/\partial z > 0$ (density increases with depth) – also $d\rho/dz$ must be sufficient to overcome the thermal expansion

- Maintenance: flush UCZ with fresh water, add salt (NaCl, $MgCl_2$) to LCZ to restore gradient, keep water clear (algae control), control evaporation.

❖Efficiency & uses

- Efficiency of heat collection ~ 15–25%; very low cost per kWh_{th}; Bhuj (Gujarat) 6000 m^2 pond.
- Uses: process heat, space heating, desalination, drying, power (ORC 25–50 kW), aquaculture.
- Others: shallow solar pond, membrane stratified pond.

7. Solar Heating & Cooling Techniques

❖A. Space heating

- Active: collectors + fluid loop + storage + radiators/ fan-coil.
- Passive (no mechanical devices):
  - Direct gain – large south windows; thermal mass floor stores heat.
  - Trombe wall – a massive dark wall behind glass with vents; air circulates by thermosiphon.
  - Sunspace / greenhouse, roof pond, isolated gain.
- Orientation, insulation, thermal mass and overhangs decide performance.

❖B. Solar cooling (refrigeration & air conditioning)

- Vapour absorption cycle ($LiBr–H_2O$ or $NH_3–H_2O$): the generator is heated by solar hot water (80–120 °C); the compressor of conventional system is replaced by absorber + pump + generator → no CFCs, silent.
  - COP ≈ 0.6–0.7 (single effect), 1.2 (double effect).
- Adsorption cooling: silica gel–water or zeolite–water.
- Desiccant cooling: solar regenerates the desiccant wheel; evaporative cooling follows.

- Solar-PV driven compressor chillers, solar ejector cooling, evaporative cooling.
- Passive cooling: wind towers, shading, roof ponds, night radiative cooling, earth tubes.

8. Solar Distillation

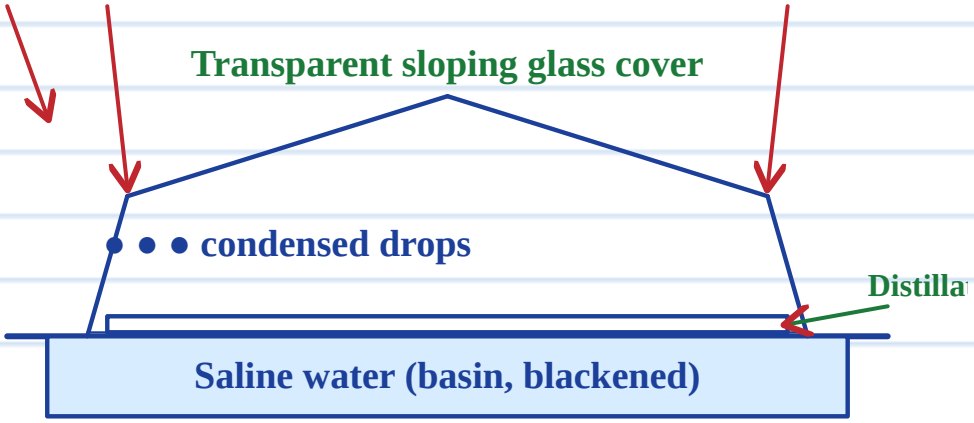


Fig 3.2 Basin type solar still

> Process of converting **saline/brackish water into potable water** using the sun's heat – evaporation followed by condensation.

- **Basin type solar still:** blackened basin with 1–5 cm saline water; sloping glass cover (tilt 10–30°); solar radiation passes the glass, heats water → evaporates; vapour condenses on cool inner glass surface, runs down to the distillate channel.
- **Productivity** ≈ 2–5 L/m²day (≈ 4 L average). Efficiency 30–40%.

$$\text{Hourly distillate } m_d = h_{ew} (T_w - T_g) A / L \text{ [kg/h]}$$

- h_{ew} = evaporative heat transfer coefficient; L = latent heat ≈ 2260 kJ/kg. Efficiency $\eta = m_d L / (A I)$.
- **Improve output:** reduce water depth, insulate basin, add wicks, reflectors, preheat feed, **multi-effect/ double basin**, use dye, cool the glass.
- Types: single-slope, double-slope, tubular, wick, multi-effect, still + FPC (active).
- Uses: drinking water for coastal/ arid villages, battery water, labs.

9. Solar Drying

- Moisture removal from agricultural products (grains, fruits, chilli, fish) using solar heated air.

- **Open sun drying:** cheap, but dust, birds, rain, loss and poor quality.
- **Solar dryers:**
  - **Direct (cabinet) dryer:** product in a glazed box, sun heats the product directly.
  - **Indirect dryer:** air heated in a solar air heater, passes through trays in a drying chamber (colour retained).
  - **Mixed mode** and **hybrid** dryers (with biomass/ electric back-up); **natural vs forced convection**; greenhouse dryer.
- **Advantages:** faster, hygienic, better colour/ quality, less loss, saves fuel.

$$\text{Moisture removed} = W_i - W_f = W_i\,(M_i - M_f)\,/\,(100 - M_f)$$

- Required energy = $m_w L$; air flow rate = $Q / (C_p \Delta T)$.

10. Photovoltaic Energy Conversion

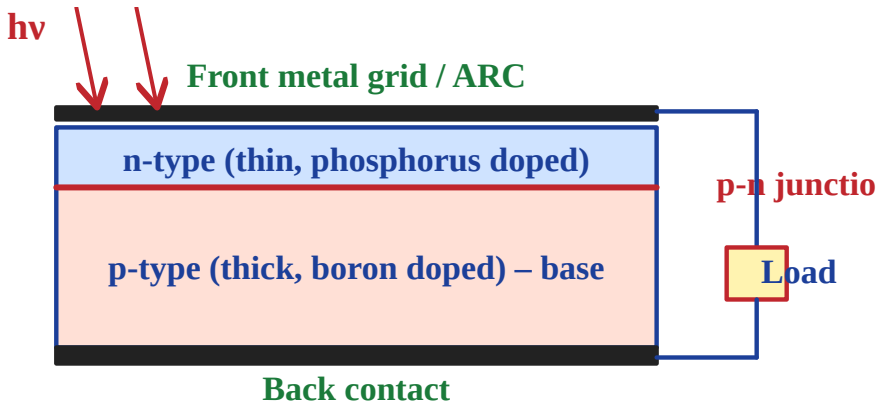


Fig 3.3 Silicon solar cell (photovoltaic effect at p-n junction)

> **Photovoltaic effect:** generation of voltage/ current when light falls on a p-n junction; discovered by Becquerel (1839).

❖Principle

- Photon energy $h\nu \geq$ band gap E_g (Si = 1.1 eV, GaAs 1.4, CdTe 1.5) creates electron–hole pairs.
- **Built-in electric field** at the junction (depletion region) separates them: electrons → n side, holes → p side → current through external load.
- Photons with $h\nu < E_g$ are not absorbed; with $h\nu > E_g$ excess is lost as heat → theoretical Si limit ~ 29–30% (Shockley–Queisser).

❖Cell characteristics

- I–V relation (ideal): $I = I_L - I_o\,[\exp(qV / nkT) - 1]$

- Short-circuit current $I_{sc} \approx I_L$; Open-circuit voltage $V_{oc} = (nkT/q) \ln(1 + I_L/I_o) \approx 0.6$ V (Si).
- Maximum power point (V_m, I_m); Fill factor $FF = V_m I_m / (V_{oc} I_{sc}) \approx 0.7–0.8$

$$\text{Efficiency } \eta = P_{max} / (I \times A) = FF \cdot V_{oc} \cdot I_{sc} / (I \times A)$$

- Standard Test Conditions (STC): 1000 W/m², 25 °C cell temp, AM 1.5.
- Temperature effect: V_{oc} falls ~ 2.3 mV/°C/cell; efficiency falls ~0.4–0.5%/°C.

❖Types of cells

Type	Efficiency
Mono-crystalline Si	18 – 23 %
Poly-crystalline Si	15 – 18 %
Thin film (a-Si, CdTe, CIGS)	8 – 18 %
Multi-junction (GaAs)	30 – 45 %
Perovskite / organic	15 – 25 % (research)

❖From cell to system

- 1 cell ≈ 0.5-0.6 V; 36 cells in series → 12 V module; modules in series (↑ V), parallel (↑ I) form array.
- System parts: PV array, charge controller (MPPT), battery, inverter (DC → AC), mounting, tracker; grid-connected (net metering) or stand-alone.
- Uses: street lights, water pumping, rooftops, satellites, solar parks, calculators, telecom towers.

II. WIND ENERGY – Sources & Potential

- Wind = air in motion, caused by uneven solar heating of earth's surface (≈ 1–2% of solar energy converts to wind), Coriolis force, and topography.
- Global patterns: trade winds, westerlies, polar easterlies, monsoons, land & sea breezes, mountain-valley winds.
- Wind power potential: world ~ 10^6 GW theoretical; India installed ≈ 50 GW+ (4th in the world); best sites need mean speed > 5–6 m/s (Tamil Nadu – Muppandal, Gujarat, Karnataka, Rajasthan, A.P. – Ramagiri, Maharashtra).

- **Advantages:** free, clean, fast installation, land can still be farmed.
  Disadvantages: intermittent, noise, bird/ bat hits, visual impact, site-specific, grid integration.

❖Wind speed variation

- **Height:** power-law $V / V_o = (h / h_o)^{\alpha}$, $\alpha \approx 1/7$ (0.14) open land, 0.25–0.4 urban/forest.
- **Weibull distribution** describes frequency of wind speeds: $f(v) = (k/c)(v/c)^{k-1} \exp[-(v/c)^k]$.
- Wind measured by **cup anemometer**, wind vane; Beaufort scale.
- **Wind rose** diagram: direction frequency.

12. Power in the Wind

- Kinetic energy of air of mass m moving at velocity V: $KE = \frac{1}{2} m V^2$.
- Mass flow through rotor area A: $\dot{m} = \rho A V$.

$$P_{wind} = \frac{1}{2} \rho A V^3 = \frac{1}{2} \rho (\pi D^2/4) V^3 \ [W]$$

- $\rho = 1.225\ kg/m^3$ (sea level, 15 °C). Power $\propto V^3$ and $\propto D^2$ – doubling wind speed gives 8 × power!
- **Power density** = $\frac{1}{2} \rho V^3\ W/m^2$ (e.g. 10 m/s → 612 W/m^2).
- Extracted power $P = C_p \cdot \frac{1}{2} \rho A V^3 \cdot \eta_{mech} \eta_{gen}$.
- Tip-speed ratio $\lambda = \omega R / V = \pi D N / (60 V)$.
- **Solidity** = blade area / swept area.

13. Horizontal Axis Wind Turbine (HAWT)

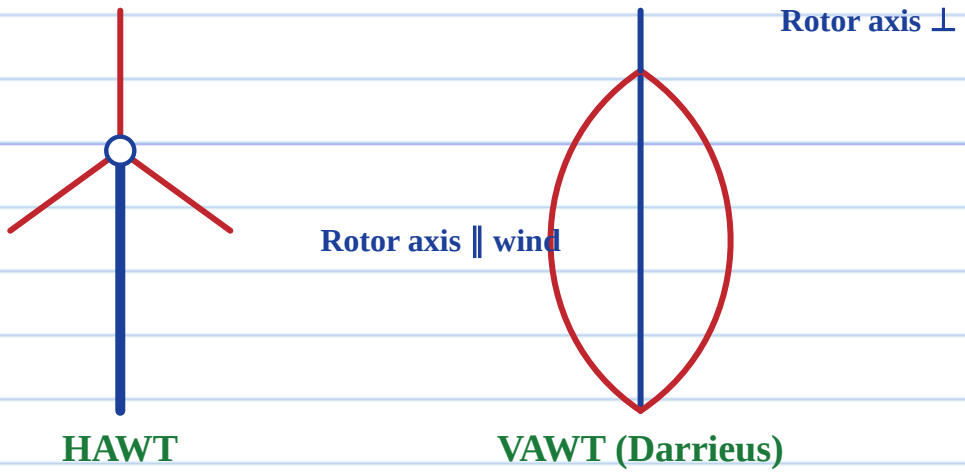


Fig 3.4 Horizontal axis (left) and vertical axis (right) wind turbines

- Rotor shaft **parallel to wind**; main components: **rotor blades** (2–3), **hub**, **nacelle** (gearbox, generator, brake, controller), **tower**, yaw mechanism to face the wind, pitch control.
- **Types:** 3-blade upwind (modern), 2-blade, down-wind, multi-blade (American farm windmill – water pumping), Dutch type windmill.
- **Advantages:** high C_p (0.40–0.50), self-starting (multi-blade), tall tower reaches steadier faster wind, pitch control.
- **Disadvantages:** needs yaw system; heavy gearbox/ generator at top; costly tower & maintenance; noise.
- **Rated wind speed** 10–14 m/s; **cut-in** 3–4 m/s; **cut-out** 20–25 m/s (feathered/ braked).

❖**Typical sizes**

- Small (< 10 kW) – farms; medium (100 kW–1 MW); large 2–15 MW (offshore).

14. Vertical Axis Wind Turbine (VAWT)

- Rotor shaft **perpendicular to wind** – accepts wind from any direction (no yaw).
- **Types:**
  - **Savonius rotor** – S-shaped drag type; self-starting, low speed, $C_p \approx 0.15$–0.3 – used for water pumping.
  - **Darrieus rotor** (eggbeater) – lift type, curved blades; high speed, $C_p \approx 0.35$; **not self-starting.**
  - H-Darrieus / Giromill – straight blades with pitch control.
- **Advantages:** generator/ gear at ground; no yaw; simple; works in turbulent wind.
- **Disadvantages:** lower efficiency, needs starting, blades see low wind close to ground, fatigue issues.

❖HAWT vs VAWT

Feature	HAWT	VAWT
Axis	Horizontal	Vertical
Yaw control	Needed	Not needed
Efficiency (C_p)	0.40–0.50	0.15–0.35
Self-starting	Yes (multi-blade)	No (Darrieus)
Generator location	Top of tower	Ground
Common use	Power plants	Small/ urban, pumping

15. Betz Criterion (Betz Limit)

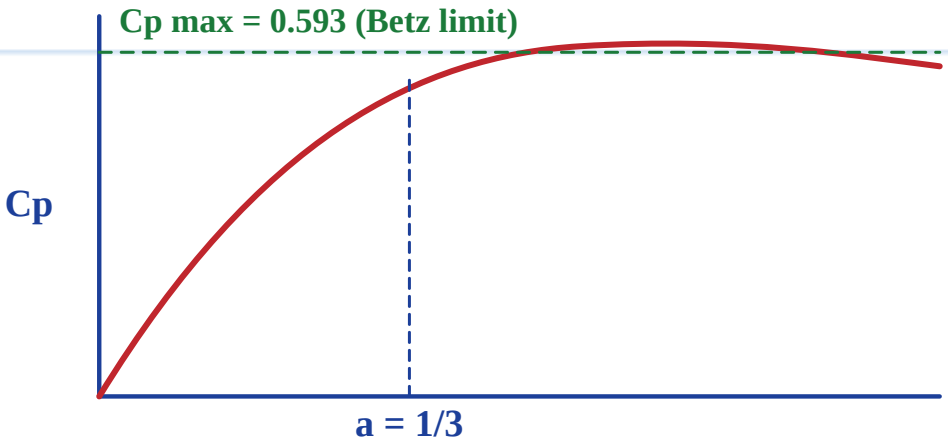


Fig 3.5 Power coefficient vs. induction factor (peak at a = 1/3)

> Betz limit: no wind turbine can convert more than 16/27 = 59.3 % of the kinetic energy of the wind into mechanical energy.

❖Derivation (momentum theory – ideal rotor)

- Wind speeds: V_1 (upstream, far), $V = (V_1 + V_2)/2$ at rotor (actuator disc), V_2 (downstream, far).
- Mass flow $\dot{m} = \rho A V = \rho A (V_1 + V_2)/2$.
- Power extracted = change in KE per sec: $P = \frac{1}{2} \dot{m} (V_1^2 - V_2^2) = \frac{1}{4} \rho A (V_1 + V_2)(V_1^2 - V_2^2)$
- Divide by available power $P_o = \frac{1}{2} \rho A V_1^3$. Let $x = V_2 / V_1$:

$$C_p = P / P_o = \frac{1}{2} (1 + x)(1 - x^2)$$

- Maximise: $dC_p/dx = 0 \rightarrow \frac{1}{2} [(1 - x^2) + (1 + x)(-2x)] = 0 \rightarrow 1 - 2x - 3x^2 = 0 \rightarrow x = 1/3$.

$$C_{p,max} = \frac{1}{2} (4/3)(1 - 1/9) = \frac{1}{2} \times 4/3 \times 8/9 = 16/27 = 0.593$$

- So at the optimum, $V = (V_1 + V_1/3)/2 = 2V_1/3$ and wake speed $V_2 = V_1/3$.
- In terms of axial induction a: $V = V_1(1 - a)$, $C_p = 4a(1 - a)^2$, max at $a = 1/3$.
- Real turbines reach $C_p = 0.35–0.50$ because of drag, wake rotation, tip losses, finite blades.
- Overall efficiency = $C_p \times \eta_{gear} \times \eta_{gen}$.

16. Performance Characteristics

- Power curve: P vs wind speed – zero below cut-in, rises as V^3 to rated speed, flat at rated power (pitch/ stall control), shuts at cut-out.
- $C_p - \lambda$ curve: C_p rises with tip-speed ratio to peak (λ_{opt} = 6–8 for 3-blade HAWT, 4–5 multi-blade slower, 2–3 Savonius), then falls.
- Torque coefficient $C_T = C_p/\lambda$; high at low λ for drag machines (good starting).
- Capacity factor (CUF) = actual energy / (rated power × 8760 h) ≈ 20–35% on land, 40–50% offshore.
- Power control: pitch control (blades rotate), stall control, yaw control, brake.
- Wind farm layout: spacing 5–9 D downwind, 3–5 D crosswind to limit wake losses.

17. Solved Problems

❖Problem 1

Wind at 10 m/s, rotor diameter 40 m, $\rho = 1.2$. $A = \pi(20)^2 = 1256.6\ m^2$. $P_{wind} = 0.5 \times 1.2 \times 1256.6 \times 1000 = 754$ kW. With $C_p = 0.4 \rightarrow P = 301.6$ kW; Betz max = 0.593 × 754 = 447 kW.

❖Problem 2 – Tip speed ratio

D = 40 m, N = 30 rpm, V = 10 m/s: $\lambda = \pi D N/(60 V) = \pi \times 40 \times 30/(600) = 6.28$.

❖Problem 3 – Wind speed at height

V at 10 m = 5 m/s, $\alpha = 1/7$. At 80 m: $V = 5 \times 8^{1/7} = 5 \times 1.346 = 6.73$ m/s; power rises by $(1.346)^3 = 2.44$ times.

❖Problem 4 – Annual energy

> 2 MW turbine, CUF = 30 % → $E = 2000 \times 8760 \times 0.30 = 5.26 \times 10^6$ kWh/yr.

18. Important Questions

- Why energy storage? Explain sensible, latent heat and stratified storage.
- Explain working of a salt-gradient solar pond with a neat sketch.
- Describe solar distillation (basin still), solar drying, solar absorption cooling.
- Explain the principle of PV conversion; I–V characteristics, fill factor, efficiency.
- Compare HAWT and VAWT; explain Savonius and Darrieus rotors.
- Derive Betz limit and show $C_{p,max} = 0.593$.

NON-CONVENTIONAL ENERGY SOURCES

Subject Code: 23A03703a | UNIT – IV : BIO-MASS & GEOTHERMAL ENERGY

1. Introduction to Biomass

Biomass = organic matter derived from plants and animals (wood, crop residue, animal dung, municipal & industrial organic waste, algae, energy crops) – stored form of **solar energy** through photosynthesis.

$$6CO_2 + 6H_2O + light \rightarrow C_6H_{12}O_6 + 6O_2$$

- Photosynthesis efficiency ≈ 1% (up to 3–5% for sugar cane).
- **Carbon neutral:** CO_2 released when burnt = CO_2 absorbed during growth.
- Sources: forest wood, agricultural residue (rice husk, bagasse, straw, coconut shells), cattle dung, urban sewage & MSW, aquatic plants (water hyacinth, algae), energy plantations (Jatropha, Pongamia, sugarcane, Subabul).
- India: ~ 500 million tonnes biomass/yr → ~ 18,000 MW potential; 12 million biogas plants possible.

❖Biomass conversion technologies

Route	Process	Product
Thermo-chemical	Direct combustion, pyrolysis, gasification, liquefaction	Heat, char, producer gas, bio-oil
Bio-chemical	Anaerobic digestion, fermentation	Biogas, ethanol
Chemical	Esterification (trans-esterification)	Biodiesel

2. Principles of Bio-Conversion

- Conversion of organic matter into usable fuels with the help of **micro-organisms (bacteria, enzymes)** under controlled conditions.
- **Aerobic digestion** (with oxygen): organic matter + O_2 → CO_2 + H_2O + heat + compost. Fast, gives **compost** and heat; no usable gas; used in aerobic composting & sewage activated sludge.
- **Anaerobic digestion (AD)** (without oxygen): organic matter → **biogas** (CH_4 + CO_2) + digested slurry (manure).

- Fermentation: sugars → ethanol + CO_2 by yeast: $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$.

❖Anaerobic digestion – 3 stages

- Stage 1 – Hydrolysis (liquefaction): *hydrolytic bacteria* (acidogens) break down complex carbohydrates, proteins & fats into simple sugars, amino acids, fatty acids.
- Stage 2 – Acid formation (acidogenesis & acetogenesis): *acid-forming bacteria* convert these into volatile fatty acids (acetic, propionic), alcohols, H_2 and CO_2.
- Stage 3 – Methanogenesis: *methanogenic bacteria* (Methanobacterium, Methanosarcina) convert acetic acid, H_2/CO_2 into methane:

$$CH_3COOH \rightarrow CH_4 + CO_2 \; ; \; CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O$$

❖Factors affecting biogas production

- Temperature: psychrophilic (< 20 °C), mesophilic (30–40 °C – best for village plants), thermophilic (50–60 °C).
- pH: 6.8 – 7.5 optimum (acid pH inhibits methanogens).
- C/N ratio: 20–30 : 1 optimum (cattle dung ≈ 24; straw ≈ 80 → mix with urine/ sewage).
- Total solids: 7–9% (dung : water = 1 : 1); retention (detention) time 30–50 days (mesophilic); loading rate 0.5–1.6 kg VS/m^3d.
- Agitation/ stirring, absence of toxic substances (antibiotics, detergents, heavy metals), seeding with active slurry.
- Particle size: smaller is better.

3. Biogas – Composition & Properties

Constituent	% by volume
Methane CH_4	55 – 65
Carbon dioxide CO_2	35 – 45
H_2S, N_2, H_2, CO	traces (< 1–2)

- Calorific value ≈ 20 – 25 MJ/m^3 (4700–5500 kcal/m^3) – vs. pure CH_4 36 MJ/m^3.

- Density ≈ 1.15 kg/m^3; ignition temperature 650–750 °C; flammability limit 5–15% in air; flame speed low (≈ 0.25 m/s); burns with clean blue flame; air needed ≈ 5.7 m^3 per m^3 gas.
- Octane number ≈ 110–130 → good for SI engines.
- Slurry after digestion = enriched manure (N, P, K retained, odour-free, weed-seed free).

4. Types of Biogas Digesters (Plants)

❖A. Batch type vs continuous type

- Batch: loaded once, sealed, emptied after gas stops (30–60 days) – used for straw/ MSW; gas production variable → needs several digesters.
- Continuous: fed daily; same volume removed; steady gas (village plants).
- One-stage vs two-stage, dry vs wet digestion.

❖B. KVIC floating drum type (India)

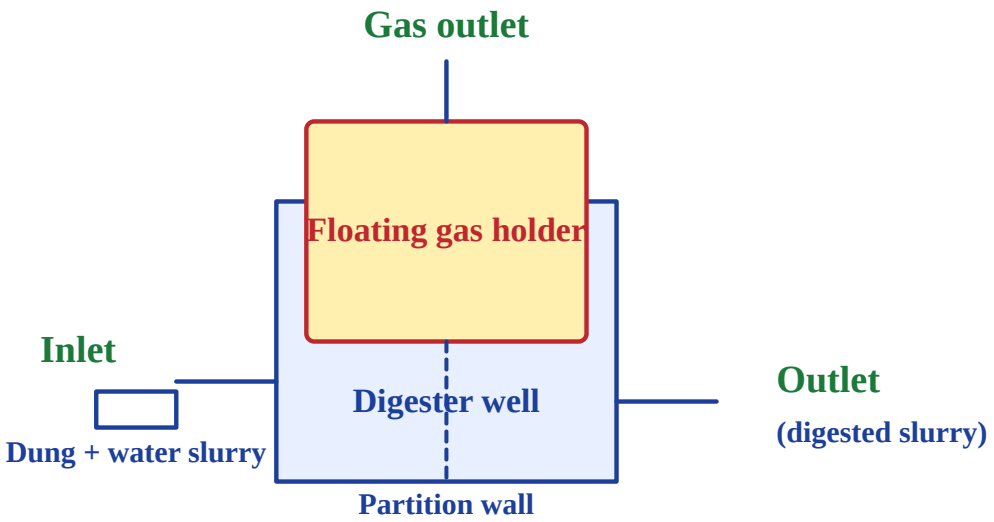


Fig 4.1 KVIC floating drum biogas plant

- Brick/ concrete digester well with central partition wall; floating steel drum gas holder moves up/down with gas storage – gives constant gas pressure (10–15 cm WG).
- Feed in through inlet pipe, slurry overflow at outlet; drum guided by central guide pipe.
- Advantages: constant pressure, volume of gas visible; easy to operate.
- Disadvantages: costly steel drum, corrosion (repaint yearly), higher maintenance, drum may stick in scum.

❖C. Fixed dome type (Chinese / Janata / Deenbandhu)

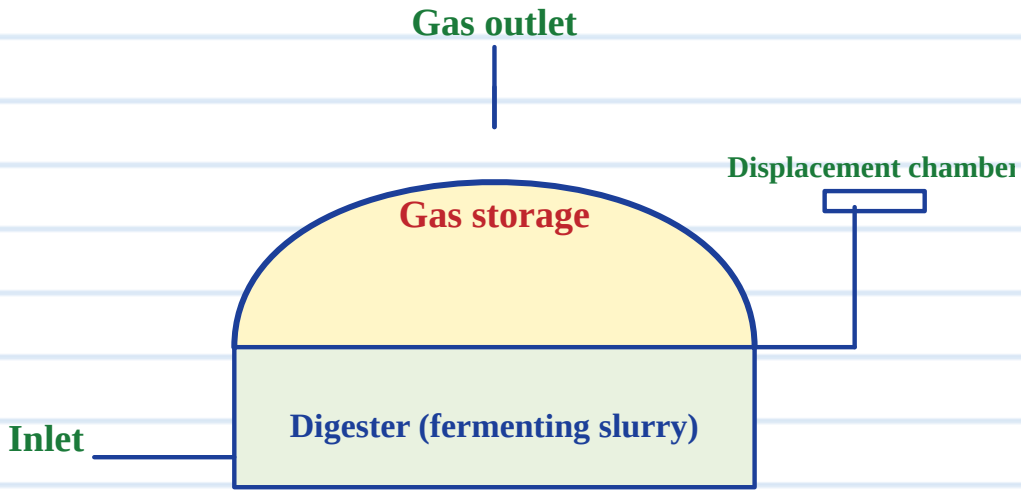


Fig 4.2 Deenbandhu / fixed dome (Chinese) biogas plant

- Underground masonry **dome** shaped digester with fixed gas holder on top; gas pressure builds as slurry is displaced into the **displacement chamber** (outlet tank) → **variable pressure** (up to 1 m WG).
- **Deenbandhu model:** spherical segments, low-cost, 1.0–2.5 m^3/day; ~ 30% cheaper than KVIC.
- **Advantages:** low cost, no moving part, long life (20 yr), underground → temperature stable, saves space.
- **Disadvantages:** gas pressure fluctuates, cracks in dome cause leakage, harder to repair, gas volume not visible.

❖D. Other types

- **Bag digester / balloon type** (PVC/ rubber bag – portable), **Plug-flow, UASB** (up-flow anaerobic sludge blanket) for industrial effluent, **Anaerobic filters, CSTR** (continuous stirred tank reactor), **Community & institutional plants, Biogas from MSW/ landfill.**

❖Comparison

Feature	KVIC (floating)	Deenbandhu (fixed dome)
Gas holder	Steel drum floats	Masonry dome, fixed
Gas pressure	Constant	Variable
Cost	High	Low (~ 30% less)
Life	8–15 yr (drum)	20+ yr
Maintenance	Painting needed	Very little
Gas leakage	Rare	Possible via cracks

5. Design & Gas Yield

- Gas yield: cattle dung ≈ 0.036 m^3/kg fresh dung (~ 0.25–0.3 m^3/kg dry VS); poultry droppings 0.07; pig dung 0.06; human excreta 0.07; water hyacinth 0.04; kitchen waste 0.1–0.2 m^3/kg.
- 1 cow/ buffalo gives ~ 10–15 kg dung/day → 0.36–0.5 m^3 gas/day.
- Daily dung input = (gas required) / (yield per kg); volume of digester = (slurry per day) × (retention time).

Digester volume $V_d = Q_{slurry} \times HRT$; Gas holder = 50–60% of daily gas

- Per-person gas for cooking ≈ 0.34 m^3/day (0.227 m^3 for lighting 1 lamp/ h).

❖Design example

A family needs 2 m^3 gas per day. Dung needed = 2/0.036 = 55.6 kg/day (≈ 5 cattle) mixed with 55.6 L water → slurry 111 L/day. Retention 40 days → V_d = 0.111×40 = 4.4 m^3 (use 4.5 m^3 plant); gas holder ≈ 1.2 m^3.

❖Typical plant sizes (India): 1, 2, 3, 4, 6, 8, 10, 15, 20, 25 m^3/day.

6. Combustion Characteristics of Biogas

- Gas is a mixture of CH_4 + CO_2; CO_2 is incombustible diluent → lower flame temp (~ 800–850 °C) vs LPG (1900 °C).
- Chemical reaction: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ + 35.8 MJ/m^3.
- Stoichiometric air: about 5.7 m^3 / m^3 (≈ 9.5 m^3 for pure methane).
- Flame speed is low (~ 0.25 m/s vs 0.45 m/s for LPG) → burner design must avoid flame blow-off; needs larger burner ports, low gas pressure, air-entrainment (primary air 60%) and orifice for high gas flow.
- Wobbe index is used to compare interchangeability of fuels; biogas ≈ 20–25 MJ/m^3.
- Purification: remove H_2S (iron oxide/ iron sponge → corrosion & SO_2), moisture (water trap), and CO_2 (water scrubbing, amine, PSA) → bio-methane (CBG) 95% CH_4 usable as vehicle fuel.
- Safety: gas leak is explosive at 5–15% in air; use ventilated room; do not use naked flame near the plant.

7. Utilisation for Cooking

- **Biogas stove:** simple burner with gas jet, air-mixing tube, burner head of 12–24 holes. Efficiency **55–60 %** (vs. wood chulha 10–15%, kerosene 40%).
- Gas consumption: 0.2–0.3 m^3/ person-day for cooking; boiling 1 L water ≈ 0.03 m^3.
- Gas pressure 75–100 mm WG required at burner; use pipes of 15–25 mm dia (GI/ PVC/ HDPE), valves, **water-drain traps**, pressure gauge.
- **Biogas lamps:** mantle lamps (0.1–0.15 m^3/h gives 40–60 W bulb light).
- **Other uses:** biogas **refrigerators**, **water heating**, **incubators**, **dual-fuel/ electricity**, **cogeneration** in dairy farms, CBG for transport.
- **Advantages over wood:** no smoke (saves women's health), no deforestation, saves time, hygienic kitchen.

8. Operation of I.C. Engines on Biogas

- Biogas is an **excellent fuel for SI engines** (high octane ~ 110–130) – can run **with petrol** (dual fuel up to 100% biogas) or fully on gas (modified carburettor/ gas mixer).
- **Diesel engines** cannot run on biogas alone (self-ignition temp 650 °C vs 250 °C for diesel) → **dual-fuel mode: pilot diesel injection (10–20%)** ignites the biogas–air mixture; saves 60–80% diesel.
- **Modifications:** gas–air mixer/ venturi in the inlet; reduce compression ratio for SI (≤ 12:1), or increase CR for gas-only SI engines; advance ignition timing (slow burning) ; corrosion-resistant materials; H_2S removal (it corrodes valves, oil).
- **Power derating:** 10–20% less power than petrol because gas displaces air (lower volumetric efficiency).
- **Specific consumption:** about **0.5–0.6 m^3 per kWh** (BHP-h ≈ 0.45 m^3); 1 m^3 gas ≈ 1.5 kWh electricity ≈ 0.6 L diesel ≈ 0.75 L petrol ≈ 0.4 kg LPG.
- **Uses:** **village power generation (genset 5–50 kW)**, water pumping, flour mills, chaff cutters.
- Engine exhaust heat can heat the digester (cogeneration).

❖Engine efficiency

$$\eta_{th} = P / (\dot{m}_{gas} \times CV) \approx 25 - 30\ \%\ \text{for gas engine}$$

9. Economic Aspects of Biogas

- **Capital cost:** 2 m^3 family plant ≈ ₹ 20,000–30,000 (subsidy through MNRE / NNBOMP ~ ₹ 7,000–12,000, extra for toilet-linked).
- **Benefits per year (2–3 m^3 plant):** saves LPG (≈ 6–8 cylinders), fire-wood (~ 3 tonnes), gives manure (~ 12 tonnes slurry; replace chemical fertilizer worth ₹ 5000+).
- **Payback period:** 3–6 years; life 15–20 years.
- **Social benefits:** clean air, women/ child time saving, sanitation, rural jobs, reduced GHG (methane capture), less deforestation.

Annual saving S ; investment C ; Simple payback = C / S (years)

❖Example

> Plant cost ₹ 25,000 (after subsidy ₹ 15,000). Annual savings: LPG 7 × ₹ 900 = 6300; fuel-wood ₹ 3000; manure ₹ 4000; total = ₹ 13,300/yr → payback = 15000/13300 ≈ 1.1 yr. Without subsidy = 1.9 yr.

- **Operating cost:** negligible – only labour for dung feeding, minor repairs.

10. Other Biomass Conversion (short notes)

- **Direct combustion:** wood/ husk boilers (efficiency 70–80%), bagasse cogeneration in sugar mills.
- **Pyrolysis:** heating biomass at 400–600 °C in absence of air → char + bio-oil + gas.
- **Gasification:** partial combustion (air-deficient, 800–1000 °C) → producer gas ($CO + H_2 + CH_4$, 4–6 MJ/m^3); downdraft gasifier for engines (1 kWh ≈ 1.2 kg wood).
- **Ethanol:** fermentation of molasses/ cane juice/ grain; blended in petrol (E20 target in India).
- **Biodiesel:** trans-esterification of vegetable oils (Jatropha/ Pongamia) with methanol.
- **Briquetting / pelletisation** of agro-waste.

II. GEOTHERMAL ENERGY – Introduction & Resources

> **Geothermal energy** = heat from inside the earth (Greek: geo = earth, therme = heat), originating from **radioactive decay** (U, Th, K) in the crust and **primordial heat** of the core.

- **Earth structure:** crust (~ 30–35 km), mantle, outer core (liquid), inner core (solid, ~ 5000–6000 °C).
- **Geothermal gradient:** average **25–30 °C per km** of depth; heat flux ≈ 0.06 W/m^2. Higher near tectonic plate boundaries, volcanoes, hot springs, geysers.
- Total heat content is huge; practical use limited to places with **high gradient** and a **fluid reservoir**.
- **Advantages:** base-load power (capacity factor 70–90%), clean, small land footprint, independent of weather, cheap running cost.
- **Disadvantages:** location-specific, high drilling cost, release of H_2S/ CO_2/ boron/ arsenic, scaling/ corrosion in pipes, possible land subsidence/ induced seismicity, need to re-inject fluid.

❖Classification of resources

1. **Hydrothermal convective systems**
   - **Vapour-dominated (dry steam):** superheated steam 180–240 °C (e.g., The Geysers – USA, Larderello – Italy).
   - **Liquid-dominated (wet steam / hot water):** water at 150–350 °C under pressure; flashes to steam when pressure drops (Wairakei – NZ, Cerro Prieto).
2. **Hot dry rock (HDR) / Enhanced Geothermal System (EGS):** hot impermeable rock at 3–5 km; water injected through one well, fractured rock heats it, returned via a production well.
3. **Geo-pressured reservoirs:** hot brine at high pressure with dissolved methane (Gulf of Mexico).
4. **Magma (molten rock) resources:** 600–1200 °C – future.
5. **Normal geothermal gradient (low-temp)** – used by **ground-source heat pumps**.

❖Resource temperature

Type	Temp	Use
High	> 150 °C	Electricity
Medium	90–150 °C	Binary power, industry
Low	< 90 °C	Space heating, spas, drying, greenhouses, heat pumps

12. Types of Wells

- **Production wells** – bring steam/ hot water up; depth 0.5–3 km (vertical, directional or slant drilled by rotary rigs with mud circulation).
- **Injection (re-injection) wells** – return spent fluid to reservoir → sustain pressure and avoid pollution/ subsidence.
- **Exploration (test) wells / slim holes** – to measure temperature, permeability, fluid chemistry.
- **Based on fluid: dry steam wells, wet (two-phase) wells, hot water wells; artesian** (self-flowing) vs **pumped** (down-hole pump).
- **Casing & cementing** – multiple casing strings protect against collapse and corrosion; **wellhead** valves control flow; **separator** removes water from steam.
- **Doublet system** (production + injection) is common for district heating and EGS.

13. Methods of Harnessing Geothermal Energy

❖(i) Dry-steam power plant

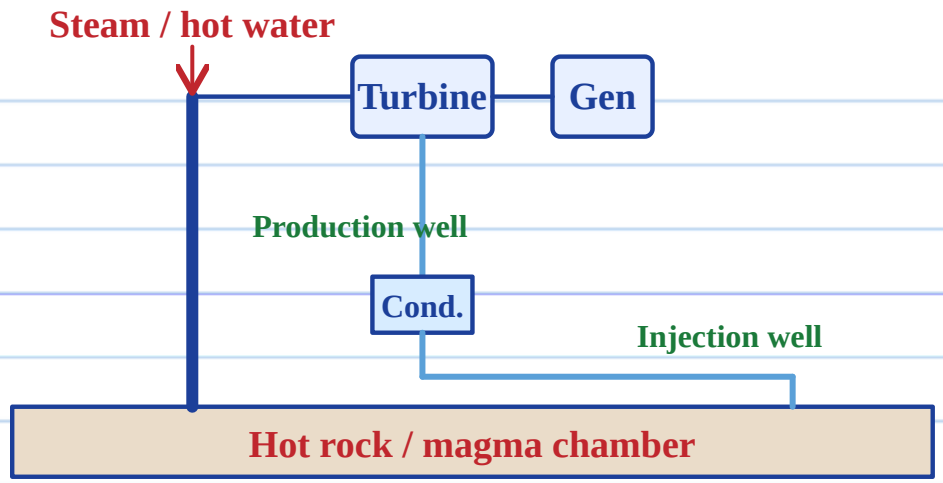


Fig 4.3 Dry-steam geothermal power plant

- Steam directly from production well (> 180 °C, 4–8 bar) → turbine → generator → condenser (cooling tower) → condensate re-injected. Simple, cheapest; e.g. The Geysers (≈ 1500 MW).

❖(ii) Flash-steam plant (single or double flash)

- High-pressure hot water (> 180 °C) is **flashed** in a separator (pressure drop) → steam goes to turbine; remaining brine re-injected or flashed a second time at lower pressure for more steam (**double flash** +15–25% power). Most common type (60% of world capacity).

❖(iii) Binary-cycle plant (ORC / Kalina)

- Geothermal water (100–180 °C) passes through **heat exchanger** to vaporise a **low-boiling working fluid** (isobutane, pentane, R-134a) in a closed **organic Rankine cycle**; geofluid never contacts the turbine and is fully re-injected → **zero emission**; suits low-temperature resources.
- Efficiency ~ 8–12%.

❖(iv) Hot dry rock (EGS)

- Water pumped down **injection well** → fracture network (hydraulic stimulation) → heated by rock → returns via **production well** → heat exchanger → power/ heat.

❖(v) Direct use (non-electric)

- **District & space heating** (Iceland – 90% homes), greenhouses, aquaculture, food drying, industrial process heat (paper, milk), spas/ balneology, snow melting, **ground-source heat pumps** (COP 3–5).

❖Thermodynamic notes

- Heat extraction: $Q = \dot{m}\,(h_1 - h_2)$; power for dry steam: $W = \dot{m}_s\,(h_{in} - h_{out})\,\eta_t$.
- Low source temperature ⇒ low efficiency (Carnot limit $\eta = 1 - T_c/T_h$; e.g. 150 °C source, 30 °C sink → 28%, actual ≈ 10–15%).
- **Dryness fraction after flashing** $x = (h_{f,in} - h_{f,sep}) / h_{fg,sep}$.

❖Flash example

> Well fluid 200 °C saturated liquid (h = 852 kJ/kg) flashed to 5 bar (h_f = 640, h_{fg} = 2109 kJ/kg): $x = (852 - 640)/2109 = 0.10$ → 10% of the mass becomes steam. For 100 kg/s well flow → 10 kg/s steam → with Δh = 700 kJ/kg across turbine, $P \approx 7$ MW (ideal).

14. Geothermal Potential in India

- ~ 340 hot spring locations, total potential estimated ≈ 10,600 MW (GSI/MNRE); no commercial geothermal power plant yet (pilot projects).
- Geothermal provinces:

Province	Examples
Himalayan	Puga & Chumathang (Ladakh) – 260 °C boiling (~ 100+ MW), Manikaran (H.P., 94 °C), Jwalamukhi
Sohana (Haryana)	Sohana
West Coast	Unkeshwar, Ratnagiri (Maharashtra), Tuwa (Gujarat)
Cambay graben	Cambay (Gujarat)
Son–Narmada–Tapi (SONATA)	Tattapani (Chhattisgarh), Anhoni (M.P.)
Godavari basin	Bhadrachalam, Manuguru (Telangana)
Mahanadi	Odisha – Atri, Deulajhari
Andaman & Nicobar	Barren Island (volcano), Narcondam

- Pilot: Manikaran 5 kW binary plant, Puga 1–2 MW project by ONGC–Iceland; Tattapani 300 kW.
- Challenges: low-grade resources, high cost, lack of data, environmental & land issues.

15. Solved Problems & Key Points

❖Problem 1 – Gas for cooking

A family of 5 needs 0.34 m^3/person/day = 1.7 m^3 gas/day. Dung = 1.7/0.036 = 47 kg/day (4 cows).

❖Problem 2 – Methane energy

100 m^3/day biogas at 22 MJ/m^3 = 2200 MJ/day = 611 kWh_{th}/day; with η = 28% → 171 kWh_e/day ≈ 7 kW average.

❖Problem 3 – Geothermal gradient

Gradient 30 °C/km, surface 25 °C: at 3 km → 25 + 90 = 115 °C; to reach 150 °C need 4.2 km.

❖Problem 4 – Carnot limit

Source 180 °C (453 K), sink 30 °C (303 K): $\eta = 1 - 303/453 = 33\ \%$ (practical 10–15%).

Key: Biogas = 55–65% CH_4, CV ≈ 22 MJ/m^3, mesophilic 35 °C, pH 7, C/N 25

16. Important Questions

- Explain the principle of bio-conversion; anaerobic vs aerobic digestion – three stages.
- With neat sketches explain KVIC floating drum and Deenbandhu fixed dome plants.
- Factors affecting gas yield; composition and combustion characteristics of biogas.
- How can biogas be used for cooking and to run I.C. engines? Economic aspects.
- Write about geothermal resources, types of wells and methods of harnessing.
- Discuss geothermal potential in India.

NON-CONVENTIONAL ENERGY SOURCES

Subject Code: 23A03703a | UNIT – V : OCEAN ENERGY & DIRECT ENERGY CONVERSION

1. Ocean Energy – Overview

- Oceans cover 71% of earth and act as a huge solar collector & store. Forms: thermal (OTEC), tidal, waves, currents, salinity gradient.
- India: coastline 7500 km; OTEC ≈ 180,000 MW; tidal ≈ 8,000–9,000 MW (Gulf of Khambhat, Gulf of Kutch, Sundarbans); wave ≈ 40,000 MW.

2. OTEC – Ocean Thermal Energy Conversion

> OTEC uses the **temperature difference** between warm surface water (25–30 °C) and cold deep water (4–6 °C at 600–1000 m depth) in tropical oceans to run a heat engine.

❖Principle & utilisation

- $\Delta T \geq 20$ °C is the minimum needed – available between ± 20° latitude (India: Lakshadweep, Andaman, Tamil Nadu coast).
- Max (Carnot) efficiency: $\eta = 1 - T_c/T_h = 1 - 278/298 = 6.7$ % (actual net 2–3 %) – very low, so large water flows needed (~ 4 m^3/s per MW).
- **By-products:** desalinated water, air-conditioning, mariculture (nutrient-rich deep water), hydrogen/ ammonia.
- Power consumed by pumps (cold-water pipe 1000 m, warm-water pump) is 20–30% of gross output.

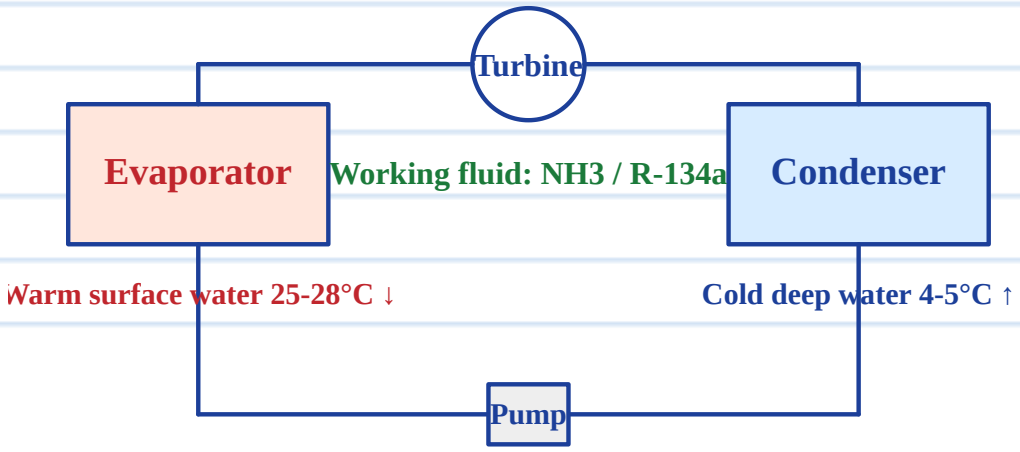
❖Setting of OTEC plants (site/ location)

- **Land-based:** shore plant near deep water; long cold-water pipe along the sea bed; stable, easy transmission, but pipe cost.
- **Shelf-mounted:** on continental shelf, fixed platform.
- **Floating (offshore):** barge/ spar/ ship anchored in deep water (e.g. India's 1 MW floating plant off Tamil Nadu 2001, Kavaratti 65 kW).
- Needs: $\Delta T > 20$ °C within short distance (< 10 km) of coast, deep water close to shore, stable sea state, near load.

- **Materials:** titanium/ aluminium heat exchangers (anti-corrosion, bio-fouling control – chlorination), HDPE/ FRP cold water pipe (dia 10 m for 100 MW).

3. OTEC Thermodynamic Cycles

❖(a) Closed cycle (Anderson cycle) – Rankine



Closed (Rankine) cycle

Fig 5.1 Closed-cycle OTEC plant

- Working fluid with low boiling point (**ammonia, propane, R-134a**) circulates in a closed loop:

1. **Evaporator:** warm sea water boils the fluid;
2. high-pressure vapour expands in **turbine** → **generator**;
3. **Condenser:** cold deep sea water condenses vapour;
4. **Pump** returns liquid to evaporator.

- Compact turbine, no direct sea water contact; heat exchangers are large.
- Net work: $W_{net} = W_{turbine} - W_{pump} - W_{sea\text{-}water\ pumps}$

❖(b) Open cycle (Claude cycle)

- **Warm sea water itself** is the working fluid. It is **flash-evaporated** in a vacuum chamber (~ 2.4 kPa, boiling at ~ 21 °C); the low-pressure steam drives a **huge low-pressure turbine**, then is condensed by cold water (direct-contact or surface) → **fresh water** produced as by-product.
- Advantages: desalinated water, no heat exchanger fouling. Disadvantages: very large turbine (low pressure), vacuum maintenance, deaeration.

❖(c) Hybrid cycle

- Open flash evaporator for steam + closed-cycle ammonia loop to generate power; gives both **power & fresh water** with a smaller turbine.

❖Comparison

Feature	Closed	Open
Working fluid	NH_3/ propane	Sea water steam
Turbine size	Small	Very large
Fresh water	No	Yes
Efficiency	2–3%	1–2%
Pressure	High	Vacuum

4. Tidal Energy

> Tides are periodic rise and fall of sea level caused by the **gravitational pull of moon (and sun)** on oceans together with earth's rotation. Two high & two low tides per ~ 24 h 50 min.

- **Spring tide:** sun, moon, earth aligned (new/ full moon) → high range. **Neap tide:** perpendicular (quarter moon) → low range.
- Tidal range for power: **> 5 m**; suitable sites: Bay of Fundy (Canada, 16 m), La Rance (France, 240 MW, since 1966), Kislaya Guba, Sihwa (Korea 254 MW), **Gulf of Khambhat (7–11 m)**, Kutch (5–8 m), Sundarbans.

❖Power from a tidal basin

Energy per tide $E = \frac{1}{2}\,\rho\, g\, A\, R^2$ (A = basin area, R = tidal range)

- Average power $P = \rho\, g\, A\, R^2 / (2\,T)$, T = tide period (12 h 25 min = 44,700 s). With efficiency η → $P = \eta \times$ above.
- Example: $A = 10\ km^2$, $R = 6$ m, $\rho = 1025$: $E = \frac{1}{2} \times 1025 \times 9.81 \times 10^7 \times 36 = 1.81 \times 10^{12}$ J/tide; $P_{avg} = 1.81 \times 10^{12} / 44700 = 40.5$ **MW** (ideal).

❖Tidal power plant types

- **Single-basin, single-effect:** one basin, one gate; generate on ebb only (or flood only) – 1 or 2 times/day, intermittent.
- **Single-basin double-effect:** generate on both ebb & flood with bulb/ reversible turbines.

- **Double-basin:** high basin + low basin with turbines in between → continuous (firm) power; costly.
- **Tidal stream/ barrage-free turbines** (marine current turbines), **tidal lagoons** (Swansea).
- **Components:** barrage/ dam, sluice gates, **bulb turbines (Kaplan type)**, generator, transformer, ship locks.

❖Advantages / Disadvantages

- Predictable, no fuel, long life, no pollution. High capital cost, intermittent, limited sites, harms marine life/ silting, corrosion of equipment.

5. Wave Energy

> Waves are created by **wind blowing over sea surface**; wave energy is concentrated form of wind (solar) energy. Typical power in deep water: $P = \rho\, g^2\, H^2\, T / (32\pi) \approx 0.5\, H^2\, T$ **kW per metre** of wave front (H = wave height m, T = period s).

- Example: $H = 2$ m, $T = 8$ s → $P = 0.5 \times 4 \times 8 =$ **16 kW/m**. Good sites 20–70 kW/m (Atlantic); India west coast ≈ 5–15 kW/m (Vizhinjam 150 kW OWC plant).

❖Wave energy conversion devices

- **Oscillating water column (OWC):** waves force air in a chamber through a **Wells turbine** (rotates same way for both air directions) → generator. (Shore/ breakwater – Vizhinjam, Pico, LIMPET).
- **Salter's duck (nodding duck):** cam-shaped float that nods with wave, converts rocking into rotation – ~ 90% efficiency in lab.
- **Cockerell raft:** hinged floating rafts; relative motion drives hydraulic pumps.
- **Pelamis (snake):** articulated cylinders; hydraulic rams – generators.
- **Point absorber / heaving buoy (Archimedes wave swing, Power buoy), wave-dragon (overtopping device):** waves overtop a ramp into a reservoir, flow back through low-head turbines.
- **Wave pump, Dam-Atoll, Pressure differential devices.**
- **Problems:** irregular waves, survivability in storms, corrosion, transmission, high cost.

6. Mini-Hydel Power Plants

Hydro plants of small capacity: **Micro** < 100 kW; **Mini** 100 kW – 2 MW (India: up to 3 MW); **Small** 2 – 25 MW. Use **run-of-river** or canal-fall/ small dam schemes. India potential ≈ 21,000 MW (SHP), ~ 5 GW installed.

$P = \eta \rho g Q H$ [W] (ρ = 1000 kg/m^3, Q = discharge m^3/s, H = net head m) $\approx 9.81 \eta Q H$ kW

- **Components:** diversion weir/ intake, **desilting tank**, canal/ **penstock**, forebay, **turbine** (Pelton – high head > 100 m; Francis – medium 30–200; Kaplan/ propeller/ cross-flow – low head < 30 m), generator, tailrace, controller (ELC).
- **Types:** low head (< 15 m), medium head (15–70 m), high head (> 70 m).

❖Economics

- Capital cost ≈ ₹ 6–10 crore/MW; low O & M, life 35–50 years, CUF 40–50%, **cost per kWh** ₹ 2–4; no fuel; gestation 2–3 years.
- Benefits: rural electrification, irrigation, flood/ no submergence, local employment; weak sides: seasonal flow, silt, site-specific, environmental flow need.

❖Example

Q = 2 m^3/s, H = 30 m, η = 0.8: $P = 9.81 \times 0.8 \times 2 \times 30$ = 471 kW; annual energy at CUF 0.5 = $471 \times 8760 \times 0.5 = 2.06 \times 10^6$ kWh.

7. DIRECT ENERGY CONVERSION (DEC) – Need & Principles

DEC = converting one form of energy **directly into electricity** without an intermediate mechanical (rotating) stage – no boiler/ turbine/ generator.

❖Need for DEC

- Conventional thermal plant: heat → mechanical → electrical; limited by **Carnot efficiency** and has many losses ($\eta \approx$ 30–40%), moving parts, pollution.
- DEC: **fewer conversion steps, no moving parts, silent, compact, reliable, low maintenance**, can reach higher efficiency (fuel cell 40–60%), useful for **space, remote & military** power.

- Exploits sources like heat, light, chemical energy, plasma.

❖Carnot cycle & its limitations

- Carnot efficiency: $\eta_c = 1 - T_L / T_H$ – maximum for any heat engine between T_H and T_L.
- Limitations in practice: need **isothermal heat transfer** (infinite area/ time), reversible processes, **no real working fluid** can follow it, low work ratio, material limits on T_H (turbine blade ~ 600–1000 °C), T_L limited by ambient (~ 300 K).
- **Fuel cells & PV** convert chemical/ light energy directly and are **not Carnot-limited** (limit set by $\Delta G/\Delta H$ or band gap).

❖Types of DEC

Device	Conversion
Thermoelectric generator	Heat → electricity (Seebeck)
Thermionic generator	Heat → electrons emitted
MHD generator	Heat/ kinetic of plasma → electricity
Fuel cell	Chemical → electricity
Photovoltaic	Light → electricity
Piezo/ Electrogasdynamic (EGD)	Mechanical/ gas flow → electricity

8. Thermo-electric Generators (TEG)

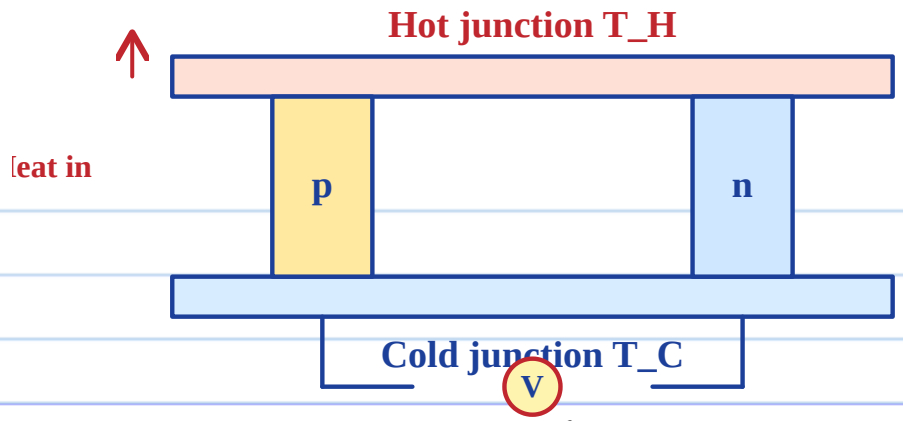


Fig 5.2 Thermoelectric generator (p-n couple, Seebeck effect)

❖Three thermo-electric effects

- **Seebeck effect (1821):** if two dissimilar conductors/ semiconductors are joined and the two junctions are kept at **different temperatures**, an **emf** is produced. $V = \alpha_{AB} (T_H - T_C)$, α = Seebeck coefficient ($\mu V/K$).

- Peltier effect (1834): reverse – when current passes through a junction of two dissimilar conductors, heat is absorbed or evolved at the junction (used in thermoelectric coolers). $Q = \Pi\, I = \alpha\, T\, I$ (Π = Peltier coefficient).
- Thomson effect (1851): heat absorbed/ released when current flows through a single conductor having a temperature gradient. $Q = \tau\, I\, dT/dx$.
- Kelvin relations: $\Pi = \alpha\, T$; $\tau = T\, d\alpha/dT$.
- Joule heating I^2R is irreversible and always present. (Note: "Joule–Thomson effect" in your syllabus refers to the cooling/ heating of a real gas when it expands through a throttle at constant enthalpy: $\mu_{JT} = (\partial T/\partial P)_h$; cooling when $\mu > 0$ (below inversion temp) – basis of liquefaction & refrigeration; $\Delta T \approx \mu\, \Delta P$.)

❖TEG operation

- Many p-type & n-type semiconductor legs connected electrically in series, thermally in parallel between hot plate (T_H) and cold plate (T_C). Heat flows from hot to cold side; carriers diffuse to cold end → voltage ≈ 0.2–0.4 mV/K per couple.

❖Efficiency & Figure of merit

Figure of merit $Z = \alpha^2 / (\rho\, k) = \alpha^2\, \sigma / k$ [1/K] ; ZT dimensionless

- α = Seebeck coeff; σ = electrical conductivity (ρ = resistivity); k = thermal conductivity.
- Maximum efficiency:

$$\eta_{max} = (T_H - T_C)/T_H \times [\sqrt{(1+ZT)} - 1]/[\sqrt{(1+ZT)} + T_C/T_H],\ T^- = (T_H+T_C)/2$$

- Larger ZT ⇒ closer to Carnot. Good materials have high σ, high α, low k ("phonon-glass electron-crystal"). $ZT \approx 1$ now (needs > 3 to compete with engines).
- Materials: Bi_2Te_3 (up to 200 °C, $ZT \approx 1$), PbTe (300–600 °C), SiGe (1000 °C – space RTGs), skutterudites, half-Heusler, TAGS, oxides.
- Power: $P = I^2 R_L$ with maximum when load R_L = internal R.

❖Applications

- Radioisotope thermoelectric generators (RTG) in Voyager, Cassini, Mars rovers; waste-heat recovery in automobiles & industry; remote pipeline/ telecom power; wearable/ body-heat devices; cooling (Peltier) – mini-fridge, CPU cooler, wine coolers; gas stove generators.
- Efficiency 5–8% – low but reliable.

9. Thermionic Converter (brief)

- A hot emitter (1500–2000 K) emits electrons by thermionic emission (Richardson: $J = A T^2 \exp(-W/kT)$) across a small gap to a cooler collector; space-charge neutralised with caesium vapour. $\eta \approx 10–20\%$.

10. MHD Generators

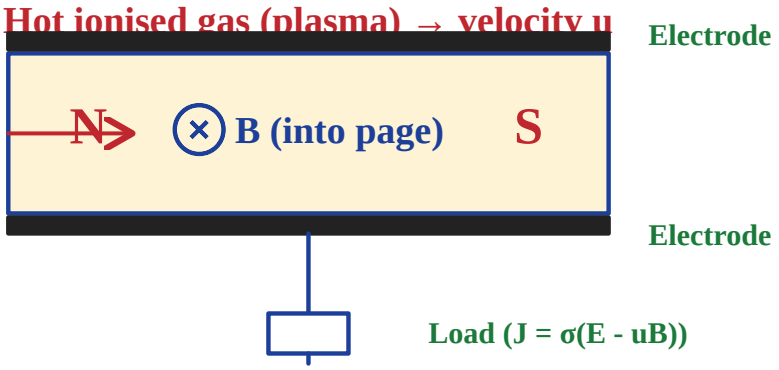


Fig 5.3 MHD generator (Faraday type) – flow, field and current mutually ⊥

> Magneto-hydrodynamic (MHD) generator converts the kinetic/ thermal energy of a hot, electrically conducting fluid (plasma) directly into electricity by passing it across a magnetic field – Faraday's law of induction applied to a moving fluid conductor instead of a copper coil.

❖Principle

- A conductor moving with velocity u perpendicular to field B gets an induced emf $E = u B$ (per unit length), acting perpendicular to both (Fleming's right-hand rule); electrodes on the channel walls collect the current → load.
- Current density: $J = \sigma (u B - E_y)$; open-circuit voltage $V_{oc} = u B d$ (d = electrode gap).
- Electrical power per unit volume = $\sigma u^2 B^2 K (1 - K)$ where load factor $K = E/(uB)$; max power at $K = 0.5$.

P_{max} per volume = $\sigma u^2 B^2 / 4$

❖Dissociation and ionisation

- At > 2500–3000 K, gas molecules **dissociate** (e.g. $O_2 \rightarrow 2O$) and then **ionise** (atoms → ions + e^-) – producing a **plasma** with free electrons (conductive).
- Combustion gases at 2500 K have very low conductivity, so the gas is **seeded** with a **low ionisation potential** alkali metal (**potassium or caesium carbonate, ~ 1% by weight**) → $\sigma \approx 10$–50 S/m.
- **Saha equation** gives degree of ionisation: $n_e\, n_i / n_0 = (2\pi m_e kT/h^2)^{3/2} \exp(-E_i/kT)$. Hence plasma needs high T, low E_i, seed.
- Non-equilibrium plasma (high electron temp) can give high σ at lower gas temp.

❖Hall effect

- In a strongly ionised gas the current J itself interacts with B: a **Hall field** E_x develops along the flow direction (**Hall effect**), because the **Lorentz force** $v \times B$ deflects the charge carriers. Hall parameter $\beta = \omega_c \tau = e B \tau / m_e$ (cyclotron frequency × collision time).
- Hall current short-circuits the axial field in a continuous-electrode (Faraday) channel, reducing output → use **segmented electrodes** (Faraday), **Hall generator** (Hall current collected by electrodes at ends), or **disc generator**.

❖Types of MHD generator

- **Faraday** (continuous or segmented electrode), **Hall**, and **Disc** (Corbino) types.
- **Open cycle:** combustion products (coal/ oil + seed) → nozzle → MHD channel (2500–3000 K) → air preheater/ steam bottoming plant → seed recovery; the gas is discharged.
- **Closed cycle:** inert gas (argon/ helium + Cs seed) circulates (T ≈ 1800–2000 K); heated in a **heat exchanger** by nuclear or coal; **non-equilibrium ionisation**; no pollution; seed not lost. Also **liquid-metal MHD** (two-phase).

❖MHD power-generation system

- **Components:** combustor/ heat source, **seeding**, **nozzle** (accelerate to ~ 1000 m/s, Mach 1–2), **MHD channel** inside superconducting/ electromagnet (B = 4–6 T), **inverter** (DC → AC), **diffuser**, air preheater, **steam bottoming cycle**, **seed recovery** unit, compressor.
- Combined MHD + steam plant efficiency **50–60 %** (MHD topping ~ 20–25%).
- **Magnetic flux** ($\Phi = B\,A$ in Weber) must be high and uniform in the channel; stronger B ⇒ more power since $P \propto B^2$.
- **Magnetic flux density** is produced by **superconducting magnet** (NbTi at 4 K) to avoid copper loss.

❖MHD accelerator & MHD engine

- If the **current is fed in** from an external source (J × B force), the **Lorentz force accelerates** the fluid: $F = J \times B$ – the **MHD accelerator / pump** (used for liquid-metal pumping in nuclear reactors, **MHD propulsion** of ships, plasma thrusters in space = **MHD engine**).
- Thrust $F = B\,I\,d$ (I = current, d = electrode gap). The machine is the **reverse of the generator** (motor action).

❖Merits & problems

- **Merits:** high efficiency (topping), no moving parts at high T, quick start-up, less pollution (SO_x locked in seed), saves water cooling, can use coal.
- **Problems:** very high temperature materials (ceramics, ZrO_2/ MgO electrodes erosion/ corrosion), strong magnets, seed recovery, nozzle erosion, low σ, voltage leakage/ arcing, high initial cost.

❖Economic aspects

- Capital ₹ is higher for a pilot MHD plant but **fuel saving 10–15%** and lower emissions give lower cost per kWh at scale; research units in India (BHEL Tiruchirappalli 5 MW pilot), Russia U-25 (25 MW), USA.

11. Electro-Gas-Dynamic (EGD) Conversion

- Charged particles (ions) are injected into a high-velocity gas stream and carried against an electric field by the flow; the gas does work on the charges → DC at high voltage (10–100 kV), tiny current. No magnetic field needed. Efficiency 5–10%. Uses: Van de Graaff-like generators, electrostatic precipitators & filtering.

12. Fuel Cells

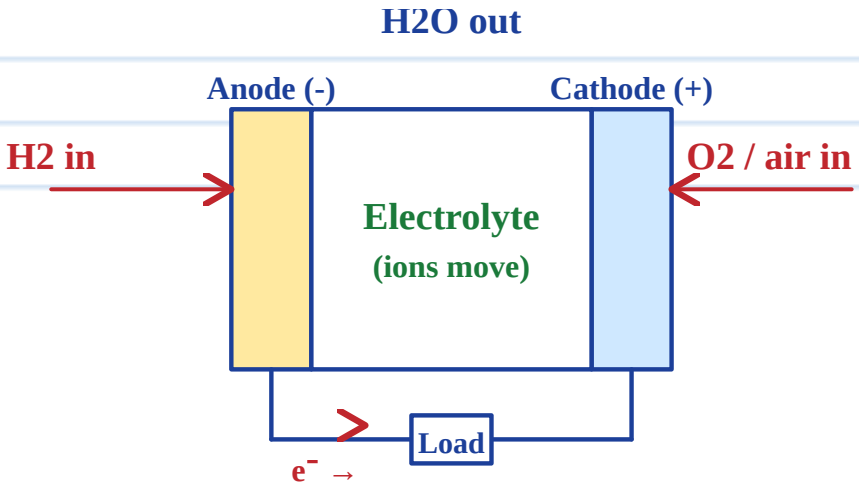


Fig 5.4 Hydrogen-oxygen fuel cell

> A fuel cell is an electrochemical device that converts the chemical energy of a fuel (H_2, CH_4, methanol) and oxidant (O_2/ air) directly into electrical energy, continuously, as long as reactants are supplied (unlike a battery which stores a limited amount).

❖Principle (H_2–O_2 acid-electrolyte cell)

- Anode (–): $H_2 \rightarrow 2H^+ + 2e^-$
- Cathode (+): $\frac{1}{2}O_2 + 2H^+ + 2e^- \rightarrow H_2O$
- Overall: $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$ + electrical energy + heat
- Electrons travel through the external circuit (load); ions through the electrolyte; catalyst (Pt) on porous electrodes speeds up reactions.

❖Faraday's laws & thermodynamic aspects

- Faraday's first law: mass of substance liberated ∝ charge passed. Second law: mass ∝ equivalent weight.

$$m = ZIt = (M/nF)\cdot It\ ;\ F = 96{,}485\ C/mol\ e^-$$

- For H_2: $n = 2$ electrons per molecule → consumption = $I/(2F)$ mol/s per cell = ≈ 1.04×10^{-8} kg/s per amp.
- Reversible (open-circuit) cell voltage: $E = -\Delta G / (n F)$. For H_2–O_2 at 25 °C: $\Delta G = -237.2$ kJ/mol → $E = 237200/(2 \times 96485) = 1.23$ V (liquid water; $\Delta H = -285.8$ kJ/mol → thermoneutral voltage 1.48 V).
- Maximum (thermodynamic) efficiency:

$$\eta_{max} = \Delta G / \Delta H = 237.2 / 285.8 = 83\ \% \text{ (at 25 °C; falls with T)}$$

- Actual efficiency $\eta = \eta_{thermo} \times (V_{actual}/E) \times$ fuel utilisation ≈ 40–60% (up to 85% with CHP).
- Not limited by Carnot since no heat engine.
- Voltage losses (polarisation): (i) activation (sluggish reaction, at low current), (ii) ohmic (electrolyte/ electrode resistance, mid current), (iii) concentration (mass transport, at high current). Operating $V \approx 0.6$–0.8 V per cell.
- Power $P = V I$; many cells stacked in series (bipolar plates) to get useful voltage.
- Effect of T & P: higher T lowers E (−0.85 mV/K) but speeds kinetics; higher P raises E: $\Delta E = (RT/2F) \ln(P_{H2} P_{O2}^{1/2}/P_{H2O})$ (Nernst).

❖Selection of fuels

- Hydrogen – best (high reactivity, clean) – by electrolysis, steam reforming, storage as compressed gas/ metal hydrides.
- Methanol (DMFC), natural gas/ methane/ hydrocarbons via internal/ external reforming, ammonia, hydrazine, biogas, coal gas (for high-T cells).
- Requirements: high reactivity at anode, availability, low cost, easy storage, no electrode poisoning (CO poisons Pt in PEMFC, S compounds poison catalysts) → purify.

❖Operating conditions & types

Type	Electrolyte	Temp	Fuel	Use
Alkaline (AFC)	KOH	60–90 °C	pure H_2	Apollo, space
PEMFC	Polymer (Nafion)	60–90 °C	H_2	Cars, portable
PAFC	Phosphoric acid	150–220 °C	H_2/ reformate	CHP buildings (200 kW)
MCFC	Molten carbonate	~ 650 °C	CH_4/ coal gas	Power plants
SOFC	Zirconia ceramic (YSZ)	800–1000 °C	CH_4, CO, H_2	Stationary, high efficiency
DMFC	Polymer	60–120 °C	Methanol	Small portable

- **Advantages:** high efficiency, silent, clean (H_2O only), modular, no moving parts, fast load response, CHP.
- **Disadvantages:** high cost (Pt, membranes), H_2 storage/ infrastructure, catalyst poisoning, durability, water management.
- **Applications:** spacecraft power & drinking water, buses/ cars (Toyota Mirai), backup & distributed power, submarines, laptops/ mobile.

❖Example

A cell gives 0.7 V at 50 A. P = 35 W per cell. For 5 kW need 5000/35 = 143 cells in series. H_2 consumption per cell = $50 \times 1.04\times10^{-8} = 5.2\times10^{-7}$ kg/s → 0.0019 kg/h per cell. Efficiency (vs. HHV 1.48 V) = 0.7/1.48 = 47 %.

13. Solved Problems

❖Problem 1 – OTEC Carnot

T_w = 27 °C (300 K), T_c = 5 °C (278 K): η_c = 22/300 = 7.3 %.

❖Problem 2 – TEG figure of merit

α = 200 μV/K, ρ = 1×10^{-5} Ω·m, k = 1.5 W/mK: $Z = (2\times10^{-4})^2/(10^{-5}\times1.5) = 4\times10^{-8}/1.5\times10^{-5} = 2.67\times10^{-3}$ K^{-1}; at T = 400 K → ZT = 1.07.

❖Problem 3 – Seebeck voltage

100 couples each α = 400 μV/K, ΔT = 150 K → $V = 100\times400\times10^{-6}\times150$ = 6 V.

❖Problem 4 – MHD power

$\sigma = 20$ S/m, $u = 1000$ m/s, $B = 5$ T, channel volume 2 m^3: $P_{max} = \sigma u^2 B^2/4 \times V = 20\times10^6\times25/4\times2 = 250$ MW.

❖Problem 5 – Fuel cell

$\Delta G = -237$ kJ/mol, $n = 2$: $E = 1.23$ V; maximum efficiency = 237/286 = 82.9 %.

14. Important Questions

- Explain OTEC; open and closed cycles with sketches; site selection.
- Tidal power: single basin and double basin schemes; derive power from basin.
- Explain wave energy conversion devices; mini-hydel schemes and economics.
- Need for DEC; limitations of Carnot cycle.
- Seebeck, Peltier, Thomson effects; figure of merit; materials & applications.
- MHD generator: principle, Hall effect, ionisation, open and closed cycles, accelerator.
- Fuel cell: working, Faraday's laws, thermodynamics, fuel selection, types.

All the Best!